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Pooling and winsorizing machine learning forecasts to predict stock returns with high-dimensional data

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ABSTRACT

We evaluate US market return predictability using a novel data set of several hundred ag- gregated firm-level characteristics. We apply LASSO, Elastic Net, Random Forest, Neural Net, Extreme Gradient Boosting, and Light Gradient Boosting Machine methods and find these models experience large prediction errors that lead to forecast failures. However, winsorizing and pooling machine learning model forecasts provides consistent out-of-sample predictability. To assess robustness, we apply machine learning methods to high-dimensional data for Canada, China, Germany and the UK as well as the Goyal–Welch data. All machine learning models we consider, except for the ensemble pooled methods, fail to significantly predict returns across our samples, highlighting the importance of pooling, evaluating additional economies, and the fragility of individual machine learning methods. Our results shed light on the sparsity versus density debate as the degree of sparsity and variable importance evolves over time.

1. Introduction

We apply an ensemble of machine learning (ML) methods to a novel high-dimensional data set of aggregated characteristics to forecast stock returns in the US, Canada, China, Germany, and the UK. Pooling winsorized forecasts from LASSO, Elastic Net (ENet), Random Forest (RF), Neural Net (NN), Extreme Gradient Boosting (XGB) and Light Gradient Boost Machines (LGBM) provide significant monthly out-of-sample R^2 (R^2_{OOS}) of 0.9 % in the US from 1975 to 2021, and 1.2 % excluding the Covid period. The ensemble's improved forecasts relative to the historical average for the US generate economic gains, measured by Sharpe ratio and utility gain. Goyal–Welch and rolling R^2_{OOS} and Giacomini and White (2006) plots illustrate that ML methods provide disparate predictability across time, while ensemble methods highlight consistent outperformance. LGBM, a relatively new ML method (Bentéjac et al., 2021), delivers R^2_{OOS} of 1.4 % in the US but is sensitive to other data sets, window sizes and time period. This sensitivity to sample, window size and time occurs for all individual ML methods we consider, and reveals both their potential fragility and the importance of considering other economies. In contrast, pooling winsorized ML forecasts generates significant predictability across samples. Our high-dimensional samples highlight that data reduction efforts such as LASSO and principal components (PC) must be met with

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caution, given substantial model uncertainty and parameter instability as the degree of sparsity, variable importance and PC compositions evolves over time.

Our paper examines the largest stock return database of financial variables (Jensen et al., 2022) shown in the cross-section to predict stock returns. We aggregate their firm-level variables to con struct 330 predictors for the US and 281–358 predictors for Canada, China, Germany and the UK. Engelberg et al. (2021) examine 140 cross-sectional predictors and find only weak evidence of out-of-sample (OOS) return predictability. Similar to Welch and Goyal (2008), their approach tests each variable individually. Yet, if extensive model and parameter instability occurs, models and predictor regressions experience episodic or pockets of local predictability, contributing to elusive OOS predictability (Timmermann, 2008; Farmer et al., 2023). Simple combination forecasts may mitigate these problems (Rapach et al., 2010); and, if there are extensive non-linearities, ML methods such as NN, RF or LGBM may uncover hidden non-linear relationships.The literature on stock return forecasting has evolved from focusing on individual predictors of financial ratios that fail out-of-sample (Welch and Goyal, 2008) to simple combination forecasts (Rapach et al., 2010) to ML methods using LASSO (Rapach et al., 2013) or multiple non-linear ML methods (Gu et al., 2020). The application of various ML methods to forecast returns raises a performance evaluation and comparison issue. For instance, in a highly cited paper on the merits of ML in predicting asset returns, Gu et al. (2020) examine multiple ML methods. They achieve monthly R^2_{OOS} of 0.39 % for a 3-layer NN, but only 0.09 % for ENet. The sharp differences in performances across models raise the questions: how does one know a priori which ML method to choose? Are 3-layer NNs always the best choice across different samples, time periods, and window sizes?

Elliott and Timmermann (2016) posit that "there is almost never a single forecasting approach that uniformly dominates all other alternatives…different forecasting models often can be combined to produce improved forecasts."[d] Hyndman (2020) suggests a divide exists in the financial forecasting community, with some aspiring to find a singular method that best describes the asset pricing process, while others view it as dynamic. The latter group assumes that the process is too complex to be modeled by a single method, as it exhibits characteristics that change over time and are often non-linear and non-stationary. This intuition motivates our approach to pool ML forecasts. Rolling R^2_{OOS} and conditional predictive accuracy (Giacomini and White, 2006) demonstrate sparse and dense models, as well as linear and nonlinear models, outperform the benchmark at different times; hence, pooling their forecasts improves predictive accuracy.

Our paper highlights the importance of examining multiple economies and ML specifications. While LGBM predicts well in the US using a rolling ten-year window, its R^2_{OOS} is insignificant using a twenty-year rolling window or recursive method and sensitive to specification in the other samples. For the UK, only RF has significant predictability across both windows; no ML methods lead to significant predictability for Canada. For the Goyal–Welch data, ENet with a twenty-year training window has an R^2_{OOS} of nearly 1 %; however, all other ML methods do not have predictability. In contrast to the mixed performance of specific machine learning methods, pooling the ten and twenty-year forecasts from these methods contributes to economically large and significant predictability across all samples. For example, ensemble methods more than double the R^2_{OOS} of Gu et al. (2020) over a similar time period (1975–2016) for the US. A rolling ten-year window of winsorized ensemble forecasts from 1975 to 2019 leads to an R^2_{OOS} of 1.28 %, a monthly return over 3 % higher than the benchmark, and a Sharpe ratio of 0.66 that is 75 % higher than the benchmark.

As the number of potential predictors rapidly exceeds the well-cited Goyal–Welch sample, the importance of handling high-dimensional data becomes relevant, raising the question of whether a parsimonious model is superior to a highly parameterized model. Giannone et al. (2021) coin the term 'illusion of sparsity,' casting doubt on methods such as LASSO, which tends to overestimate the degree of sparsity when the number of true regressors in the data-generating process is large. They show sparse models suffer from randomness in selecting the right variable in high-dimensional settings with high collinearity between variables and when the effect of factors are weak. Li et al. (2018) and Fan et al. (2023) 'embrace the blessing of dimensionality' and argue to use as much data as possible since it contains more information. Kelly et al. (2023) argues for the 'virtue of complexity' in return predictions, highlighting the importance of heavily parameterized nonlinear models. These papers ignore concerns about unstable parameters and evolving model performance over time, aiming to find optimal models, not test dynamic return predictability.

We evaluate the time-varying performance of LASSO, Ridge, and ENet procedures as well as factors that determine their performance. Results show that ENet outperforms LASSO and Ridge more often, and all three approaches outperform the benchmark at different times. Periods of high volatility are related to larger magnitudes of shrinkage and higher predictability of ENet. We find the optimal regularization parameter for ENet changes over time, implying the optimal level of sparsity evolves. More dense models are preferred when volatility is high. We investigate the effect of parameter instability on PC and PLS, methods that embrace dense methods. For the US, the number of PC required to explain 90 % of the variance substantially varies over time and factor loadings experience large, rapid shifts; Breitung and Eickmeier (2011) show the consequences of structural breaks in the factor loadings. In our data of 330 variables, PLS takes approximately 100 components to explain 50 % of the variance and experiences large rotational shifts. Results reveal both PC and PLS do not provide consistent parameters over time, can not succinctly summarize the variance over the entire sample, and fail to forecast returns significantly. The sparsity vs. density debate is difficult to resolve without examining the impact of parameter instability.

[d] In their review of the M4 forecasting competition, Petropoulos and Makridakis (2020) find all top performers combined forecasts from multiple models, and sophisticated ensemble forecasts are superior to individual methods (see also, Montero-Manso et al. (2020)). A recent survey of fifty years of combination forecasts notes that combining multiple individual forecasts dates back at least a hundred years ago to Galton, and combining multiple forecasts is often a better approach than identifying a single best forecast (Wang et al., 2022).

Nonlinear ML methods, parameter instability, short windows, and high-dimensional data potentially pose forecast challenges since they may generate extreme predictions, forecast failures, and negative R^2_{OOS}. To mitigate large forecasts, we recursively winsorize model forecasts and show that both ML and ensemble methods benefit from relatively large winsorization; shorter windows and nonlinear methods particularly require high levels of forecast winsorization. The intuition behind forecast winsorization is similar to the motivation behind trimming forecasts when you have multiple forecast inputs. Stock and Watson (1998) argue for trimming of their 49 different forecast methods to avoid 'crazy' forecasts; Jose and Winkler (2008) find that winsorizing combined forecast methods *across individual variable forecasts* lead to robust forecasts and boosts predictability by 15–45 %, where the optimal degree of winsorizing depends on the variability of the individual forecast.

In our application of aggregate data, we show that winsorization of outliers considerably "shrinks the forecasts towards the historical mean" and achieves "forecast stabilization" similar to combination forecasts (Rapach et al., 2010). They show individual regression forecasts are "often highly variable" or "unrealistically large values for the equity premium," and "individual forecasts appear to contain substantial noise". We show that winsorization also achieves these goals by shrinking the forecasts *across time* in all five economies and the Goyal–Welch data. We further document that winsorization of ML forecasts outperforms shrinkage (Li et al., 2018; Zhang et al., 2020), which reduces the variance of the forecasts, but insufficiently as the forecasts sometimes are very large. Finally, we find that simple pooling methods outperform ensemble methods that heavily weight recent superior model performance. This supports Elliott and Timmermann (2016) who recommend equal-weighting model forecasts due to endemic model misspecification, model instability, and parameter instability that imply no model is correct all the time.

Overall, our paper examines three relatively novel issues from return predictability, pooling winsorized ML forecasts to multiple high-dimensional data sets. The rest of the paper is structured as follows. Section 2 explains the data and methodology. Section 3 presents the out-of-sample results, including sections that consider alternative winsorization levels, pooling methods, and samples. Section 4 takes a deeper dive into explaining the success of pooling, the sparsity vs. density issue, and parameter instability's impact on dimensionality reduction. Section 5 concludes.

2. Data and methodology

2.1. Data

Jensen et al. (2022) provide long-short returns for hundreds of factors to enable replication of cross- section determinants of stock returns. In contrast to Dong et al. (2022) who use long-short returns, we are interested in the predictor variables that create these portfolios, and their online appendix contains detailed instructions to download firm-level data from WRDS. Because most countries have few firms or a short-time sample, we select countries with relatively high numbers of firms and at least twenty years of data. These include the USA, Canada, Germany, China, and the UK. Inspection of the data highlights that some variables suffer from data integrity issues because they exhibit extreme outliers in both directions. This has severe adverse effects on model fit and forecasting. To mitigate outliers, a recursive winsorization process is adopted that cuts off all variable values at the 2 % and 98 % levels, using only data available in real time. This winsorization process is similar to the common approach applied to minimize data outliers and differs from winsorizing forecasts that we also use to moderate forecast outliers. All independent variables are standardized, which improves the efficacy of some methods.

Appendix I clusters the variables in categories and lists an example for each category, as well as the maximum number of variables per category across countries and time; a full data of all fields and information can be found at Brian Kelly's Github replication website. Most variables are financial ratios scaled by assets or market capitalization. Before we aggregate the firm-level data to the country level, we follow the literature (e.g., Fama and French, 2008; Hou et al., 2015; Giannone et al., 2021; Engelberg et al., 2021; Dong et al., 2022) and remove micro level firms in the bottom 20th percentile of firm market cap. The monthly firm variables are then aggregated using a capitalization-weighted approach following Jensen et al. (2022) that mitigates overweighting of large firms. We determine each firm's simple market capitalization weight and then readjust the weights of the largest 20 % of firms to the 80th percentile weight.

Returns are calculated as the traditional value-weighted average of the non-micro cap firms, without the capped weight adjustment, to simulate an investment in the market. We follow Fama and French (2015) and aggregate variables when there are at least thirty firm observations in that time period. We only use data available in real-time to the forecaster in the OOS procedures. Financial reporting time frames typically vary between 1 and 3 months after the quarter's end. To be prudent, we lag variables by four months following how the variables are calculated by Jensen et al. (2022).

Table 1
Summary statistics for countries.

Country	Trunc. start date	Num. assets	Num. variables	Ann. mean (%)	Min (%)	Max (%)	Vol. (%)	Sharp ratio
USA	1954:01	35,573	330	7.909	−23.25	14.367	15.86	0.499
Canada	1984:07	6,565	358	9.471	−27.503	20.537	19.585	0.484
UK	1992:02	6,371	308	5.143	−18.078	15.268	16.353	0.314
Germany	1994:05	1,882	307	6.05	−13.836	16.01	16.602	0.364
China	1998:05	4,759	281	7.646	−11.589	17.81	21.23	0.360

The table reports summary statistics for one-month excess returns of individual countries in the data set.

Table 1 presents the data sets across five countries, ranging from 281 predictors in China to 358 in Canada. Although data might be available for a long time frame, it does not always contain all predictors. Manual data inspection indicates cut-off points where multiple additional variables become available and serve as a starting point for the in-sample period, indicated by the 'Truncated start date' column. The table gives summary statistics for excess returns in each country, including the number of assets used to calculate the respective market returns. The US market has the largest number of assets with 35,573, more than five times as many assets as the next country, Canada, at 6,565. The table also presents mean return, minimum and maximum returns, volatility, and Sharpe ratios, however direct comparisons across economies can be misleading due to different time periods.

Since we have 330 variables, we do not present summary statistics for each of them or a correlation matrix 330×330, instead Fig. 1 presents a histogram of bivariate correlations of the variable space for the USA. It highlights only moderate collinearity between the variables, which is surprising given that financial ratios often comove together, with only about 3.5 % of the bivariate correlations above 0.5 and 1.7 % below $-.5$.

2.2. Methodology

Effective ML estimation requires guarding against model overfitting to the training set since this leads to spurious fit measures with poor dynamic performance. We apply cross-validation to determine optimal values of hyper-parameters for regularization, model complexity, and learning rates as part of the model training process. This approach trains models multiple times with different hyper-parameters on a split sample and validation set and selects the hyper-parameter values with the best fit measures. Each ML method possesses differences in how hyper-parameters are conceptualized, and we briefly describe these below. The generalized functional form is:

$$r_{t+1} = g_{t+1}(X_t) + \epsilon_{t+1} \tag{1}$$

where X_t is the vector of predictor variables and time $t = 1, \ldots, T$ and g_{t+1} is the estimator function returning the conditional expected returns at time $t + 1$. We allow the estimator function to evolve over t, which allows the parameters to capture dynamic relationships. The following sections briefly walk through the hyper-parameter tuning process and then describe the different methods used in the forecasting approach.

2.3. Hyper-parameter tuning

For each ML method, we tune hyperparameters at an interval of six months, where the six-month interval compromises between computational time and method accuracy. Experimental execution shows that tuning every six months tends to be more accurate than training annually and similar to monthly tuning. We use a time series walk-forward cross-validation method that cuts the sample set into five different splits, the first split consisting of half of the observations. for assessment with a validation set consisting of the subsequent one-tenth of the observations. The second to fifth split each add an incremental one-tenth of observations to the training sub-set, and uses the next one-tenth of the observations for assessment. The method preserves the temporal structure of the data while allowing for cross-validation.

The tuning framework minimizes the mean squared error in cross-validation. Given a very large tuning parameter space, particularly for complex ML algorithms, grid search is impractical and we apply a Bayesian Tree-Structured Parzen Estimator (TPE) algorithm (Bergstra et al., 2011) to select the most likely parameter combinations dynamically. The TPE Method in Optuna (Akiba et al., 2019) is a hyperparameter optimization algorithm based on Bayesian optimization. It models the performance of hyperparameters using two probabilistic distributions: one for parameters associated with good results and another for all trials. By focusing on promising regions of the search space, TPE efficiently guides the selection of hyperparameters, especially in complex, hierarchical spaces, optimizing the search process to find the best configurations with fewer trials. Up to 50 trials with different hyperparameters

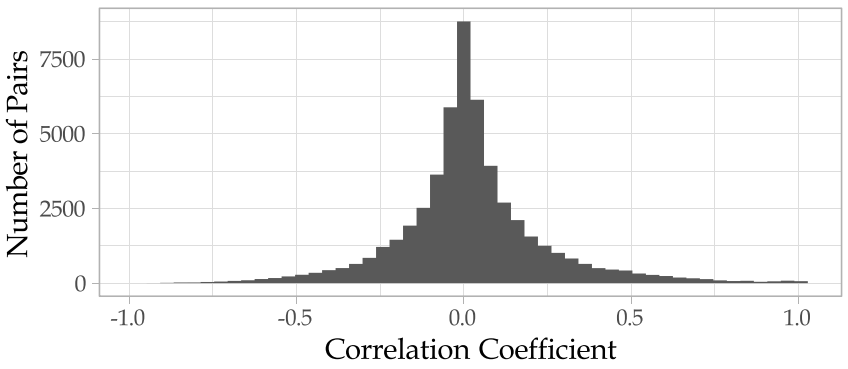


Fig. 1. Correlations between independent variables.
The figure shows a histogram of the top half of the correlation table of all independent variables.

are considered to find an optimal parameter set with median pruning, which stops a trial if its best result is worse than the median of previous trial results for performance optimization.

2.4. Linear methods

The first class of methods we consider uses linear combinations of predictors. Following Rapach et al. (2010), we adopt a simple combination forecast method that averages forecasts of individual regressions of all variables to forecast market returns. We incorporate this model as part of our ensemble. Since there are a large number of variables, it is useful to perform variable selection and/or shrinkage methods. These methods penalize linear regression models by iteratively shrinking parameter coefficients to the point where only the most important predictors remain (Gu et al., 2020). Three common methods are ENet, Least Absolute Shrinkage and Selection Operator (LASSO), and Ridge, where LASSO and Ridge are specific forms of the more general ENet specification. Tibshirani (1996) introduced LASSO as a predictor shrinkage and variable selection method that shrinks variable coefficients to zero, so that only the relevant variables driving the variation in the outcome remain. In contrast, Ridge does not perform variable selection but instead shrinks coefficient estimates. ENet (Zou and Hastie, 2005) aims to mitigate LASSO's issue of arbitrarily selecting a random regressor from a list of correlated regressors and Ridge's issue of retaining all the regressors with shrunken parameter estimates. Successful applications of LASSO and ENet for forecasting returns include Rapach et al. (2013), Rapach and Zhou (2020), and Zhang et al. (2020).

ENet, LASSO and Ridge share a mathematical foundation, and they differ by an L1-regularization. If L1=1, selection (LASSO) is applied, and if L1=0, then strict shrinkage is applied (Ridge). If $0 < L1 < 1$, ENet is estimated. The methods have a second common parameter, α that determines the magnitude of the penalty. We find that setting a static α value performs less well than dynamically estimating the optimal magnitude through hyperparameter tuning. Thus, we define an α search space between 10^{-5} to 10^{5} and evaluate walk-forward cross-validation for optimization. We then leverage the mean of the α value with the smallest MSFE and the highest α value that has an MSFE that is less than one standard error above the smallest α as our parameter for the estimation. This adjustment is applied to avoid overfitting to a sample-specific α.

2.5. Nonlinear methods

While classical asset forecasting methods tend to assume a linear and sparse model space, research reveals that the sparsity and linearity assumptions might be suboptimal (e.g. Kelly et al., 2023) and incorporating nonlinearities improves prediction. We consider several nonlinear models including NN, RF, XGB and LGBM. The latter three tree-based decision methods allow the inclusion of multi-way interactions and higher-order terms nonparametrically. They create decision trees of combinations of predictors and then aggregate these predictions. However, a drawback is that they are prone to overfitting and require regularization.

Gradient boosting (Friedman, 2001; Breiman, 1997) sequentially creates many simple decision trees and aggregates them in an ensemble to gain predictive power. As the simplicity of trees can lead to poor fit, boosting is a mechanism that incorporates information on the error of prior trees to help improve the fit of subsequent trees. Friedman (2002) suggests applying stochastic gradient descent to make gradient boosting more robust, which is incorporated in the XGB (Chen and Guestrin, 2016) algorithm we use. When fitting the XGB method, we apply hyperparameter tuning to the number of trees (100 to 1000), the minimum weight of a child node (10 to 1000), the maximum tree depth (4 to 12), the learning rate (0.005 to 0.05), L1 regularization (meaning selection within this method, 1^{e-8} to 10), L2 regularization (meaning shrinkage, 1^{e-8} to 10) and minimum gain for additional trees to be added (1^{e-8} to 10).

LGBM, an algorithm developed by Microsoft in 2017 (Ke et al., 2017) differs from XGB by growing trees leaf-wise instead of top-down, choosing the leaves with the highest loss delta at each step. While this algorithm is rapid, a balance needs to occur between in-sample accuracy, supported by tree depth and number of leaves, and overfitting. We tune the number of trees (100 to 1000), the number of leaves (2 to 256), L1 regularization (1^{e-8} to 10), L2 regularization (1^{e-8} to 10), learning rate (1^{e-5} to 1^{e-1}) and the minimum number of samples at each child leaf (5 to 100).

Instead of gradient boosting, RF (Breiman, 2001) uses bootstrap aggregation of decision trees created by bootstrapping the observations and randomly selecting model predictors. This partitions the space of available predictors and calculates a score for the outcome variable in each partition. To achieve robust predictability, these forecasts are then combined by a voting mechanism based on the CART algorithm following Breiman et al. (1984). When calculating the RF models, we apply hyper-parameter tuning to the number of trees in the forest (100 to 1000), the minimum number of predictor observations needed to split an internal node (3 to 10), the minimum number of target observations in a leaf node (3 to 10), the maximum depth of each tree (3 to 10) and the maximum number of predictors to consider for a split (5 to 20).

NNs are a ML method designed to incorporate complex interactions in the model specification. We apply the Keras Python package (Chollet et al., 2015) to implement our NN since it allows for a great level of flexibility in designing and tuning the hyper-parameters required for this complex procedure. We tune the number of hidden layers (1 to 5), and set the neurons in each layer of the NN using an inverse geometric pyramid rule outlined by Masters (1993) and Gu et al. (2020), with up to 128 neurons in the first hidden layer, 64 in the second, and 32, 16 and 8 neurons in the third, fourth and fifth layers respectively. We set the loss function to MSFE, and tune the activation function (relu or leaky-relu) and the learning rate (1^{e-5} to 1^{e-1}) and the dropout rate (0.2 to 0.5).

2.6. Ensembles

We apply two different methods for ensemble creation. The first is a simple averaging approach of the two rolling windows (120 and 240-month) with three permutations, the simple mean, median and trimmed mean, with 10 % symmetric trimming. The trimmed mean and median are less susceptible to outlier forecasts.

Our second ensemble method weights model forecasts of the ten and twenty-year rolling windows based on information on their prior performance. Stock and Watson (2004) show it can be beneficial to 'tilt' combining weights towards models with lower mean squared forecast errors, which places greater weight on more recent performance. In contrast to their paper that weights individual predictors' performance, our ensemble method places greater weight on machine learning models with lower MSFE and greater weight on recent performance by using a discount rate of 0.95. For robustness, we also assess discount rates of 0.90 (high discount), 0.99 (low discount) and 1.0 (no discount). Raftery et al. (2010) introduced dynamic model averaging (DMA), a recursive method that allows for both the coefficients and the model weights to change over time using a combination of forgetting factors. It uses two coefficients for each variable, one indicating the variable's coefficient in the model and the other indicating binary inclusion in the model. The method utilizes approximated Markov switching models to update the probabilities of the models themselves, as well as the coefficients in each model over time. DMA has shown promise in forecasting inflation by simultaneously addressing both parameter instability and model uncertainty (Koop and Korobilis, 2012; Nonejad, 2021).

3. Results

3.1. Out-of-sample results and model uncertainty

Table 2 presents the results for the ML methods applied to the USA data set, using ten and twenty-year rolling windows. The historical average (HA) benchmark is formed using a recursive window. We calculate R^2_{OOS} following Welch and Goyal (2008) and use the corresponding Clark-West (CW) statistics (Clark and West, 2007) to evaluate the significance of the R^2_{OOS} metric relative to the HA benchmark. The data begin in 1954:01, and the OOS period in 1975:01. Recursive and rolling SCF are reported as additional references in the second row and third rows. Column I presents the R^2_{OOS} for the full sample and does not winsorize the forecasts. The recursive and rolling SCF240 do not yield significant predictability, and the shorter SCF120 window has a large negative R^2_{OOS}; the numbers behind the model acronyms indicate the size of the training window. Linear methods such as LASSO and ENet have R^2_{OOS} less than zero.[e] LGBM120 is the only ML procedure that delivers predictability at the 95 % confidence level; in contrast, LGBM240 does not possess significant R^2_{OOS}, nor does a recursive approach presented later in Table 12. Column II winsorizes the model forecasts at 10 % and presents the R^2_{OOS} over the full sample. Winsorizing forecasts mitigates the impact of extreme predictions potentially caused by model overfitting, particularly for nonlinear ML methods and relatively short windows. The SCF120, ENet120, and XGB120 are now significant at 95 % and provide strong evidence that winsorizing forecasts raise predictability. However, using a longer window, the ML methods, except for LASSO, are insignificant. A shorter window better captures evolving parameters but may experience sharp forecast failures due to overfitting, especially given high-dimensional and potentially nonlinear data. In the next section, we show that shorter windows have more extreme forecasts mitigated by winsorizing.

Columns III and IV present results excluding the Covid period for non-winsorized and winsorized forecasts at 10 %; the 2020–2021 period was an exceptional event, and most papers that examine return predictability are written pre-Covid and do not include this period. Predictability increases for SCF120, ENet120, LGBM120 and XGB120 with winsorized forecasts. Columns V and VI present nonwinsorized and winsorized forecasts for 1987–2016, the same period as Gu et al. (2020), to provide a comparison to their results. LGBM120 and XGB possess an R^2_{OOS} exceeding 1 % for the winsorized forecasts for both periods, and SCF120 is also significant at 95 %. NN is not significant in contrast to their results.

The bottom three rows of Table 2 pool the forecasts from the six ML methods and SCF across the two rolling windows. Ensemble predictability in column I (2021W0) ranges between 0.67–0.78 %, and winsorizing boosts the forecast to 0.76–0.82 %. Excluding Covid, ensemble R^2_{OOS} range from 0.97 to 1.02 % using winsorized forecasts and 1.08 −1.15 % for the period examined by Gu et al. (2020). Their monthly predictability for NN1-NN5 ranged between 0.35–0.39 %. Hence, a combination of applying high-dimensional data, pooling ML methods, and winsorizing the forecasts leads to approximately three times the R^2_{OOS}. Ensemble predictability also leads to meaningful economic gains. For the ex-Covid period, returns increase from 6.14 % annually for the HA to 7.92–8.30 % for the ensembles. Sharpe ratios rise from 0.38 for the HA to 0.46–0.48 for the three ensemble methods forecasts using a 10 % winsorization level. Utility gains range from 1.20 to 1.57 % for the three ensemble methods and the Sortino ratio increases from 0.64 for the HA to 0.76–0.80.

[e] Note, we also tried Adaptive Lasso (AdaLasso) methods following Zou (2006) and Medeiros and Mendes (2016). They show the AdaLasso consistently chooses the relative variables as the number of observations increases. In our case, this procedure has similar results to Lasso, e.g., the R^2_{OOS} was −2.60 % for the full-sample and −.363 with 10 % winsorization. Following Medeiros and Mendes (2016), we also tried choosing our hyperparameters using BIC critiera and find little change in inference.

Table 2
USA R^2_{OOS} Results with winsorization levels, ex-Covid comparison and investment performance metrics.

Model	Training Window	I 2021W0	II 2021W10	III 2019W0	IV 2019W10	V 8716W0	VI 8716W10	Return	Sharpe	UGain	Sortino
HA								6.141	0.377	–	0.637
SCF	REC	0.017	−0.088	0.037	−0.077	−0.387	−0.096	8.027	0.458	0.046	0.765
SCF	120	−35.043	0.476 **	−37.98	0.533 *	−55.611	0.401 **	6.672	0.396	0.296	0.662
LASSO	120	−2.052	−0.009	−1.919	0.199	−2.567	0.01	6.169	0.387	0.109	0.644
ENet	120	−0.072	0.529 **	0.216	0.785 **	−0.375	0.398	7.163	0.477	1.212	0.779
LGBM	120	1.446 ***	1.226 ***	1.627 ***	1.401 ***	2.942 ***	1.564 ***	6.227	0.426	0.491	0.698
RF	120	−2.611	−1.303	−2.236	−0.912	−1.628	−0.513	7.758	0.572	2.154	0.930
XGB	120	−0.089	0.814 ***	0.033	1.019 ***	0.363	1.362 ***	6.644	0.409	0.437	0.682
NN	120	−8.016	−1.048	−8.442	−1.227	−7.281	−0.787	5.469	0.348	−0.433	0.588
SCF	240	−0.012	−0.093	0.041	−0.078	−0.359	0.012	8.027	0.458	0.000	0.765
LASSO	240	0.101 *	0.499 **	0.067	0.441 *	0.254	0.512 *	8.018	0.459	0.007	0.766
ENet	240	−0.709	−0.22	−0.634	−0.324	−1.457	−0.739	8.283	0.484	0.339	0.802
LGBM	240	−0.13	−0.238	−0.059	−0.131	0.336	0.144	7.753	0.444	−0.220	0.741
RF	240	−3.153	−2.189	−3.056	−2.032	−3.118	−1.738	6.705	0.467	−0.284	0.764
XGB	240	−0.086	−0.032	−0.157	−0.148	−1.067	−0.702	7.942	0.490	0.311	0.806
NN	240	−1.322 *	0.096	−1.451	0.062	−1.367	0.018	7.620	0.454	−0.137	0.764
ENSMEAN		0.748 **	0.796 ***	0.937 **	1.003 ***	0.999 **	1.126 ***	8.287	0.481	1.570	0.798
ENSMED		0.668 **	0.764 ***	0.876 **	0.97 ***	0.923 **	1.079 ***	8.027	0.458	1.261	0.765
ENSTMEAN		0.775 **	0.821 ***	0.964 **	1.021 ***	1.024 **	1.145 ***	7.917	0.455	1.201	0.757

The table presents R^2_{OOS} and economic gains of the ML models relative to the historical average (HA) benchmark, which uses a recursive (REC) window. We also present a simple combination forecast (SCF). The ML models apply rolling windows of 120 and 240; the ensemble averages both rolling windows. * indicate that the reduction in MSFE is significant according to Clark-West tests. Column one has an out-of-sample (OOS) period of 1975:01–2021:12, and uses no winsorization, W0. Column II winsorizes the forecasts by 10 %, W10. Columns III and IV use OOS from 1975:01–2019.12, and hence exclude Covid. Columns V and VI (8716) present the 1987:01–2016,12 period that coincides with Gu et al. (2020). The last three columns present the annualized mean return, Sharpe ratio, and Utility Gain. ENSMEAN is the ensemble simple mean of forecasts, ENSMED pools forecast using the simple median, and ENSTMEAN ensembles using a 10 % trimmed mean.

*** $p < 0.01$.
** $p < 0.05$.
* $p < 0.1$.

3.2. Forecast winsorization

Table 3 investigates the sensitivity of different forecast winsorization levels of 0.025, 0.05, 0.15, 0.20, and 0.25. We compare these results to an alternative procedure that shrinks the forecasts to the HA (Lin et al., 2018; Zhang et al., 2020). Results show that winsorization affects the predictability of several ML methods. Winsorizing forecasts steadily raise the R^2_{OOS} for both ENet120 and XGB120 from 0.2 % and 0.1 % to 1.4 % and 1.4 %, respectively. Winsorizing forecasts mitigates large forecast breakdowns, particularly for the shorter rolling window; e.g., NN120, RF120 and LASSO120 have R^2_{OOS} of −8.44 %, −2.24 % and −1.90 %, but winsorizing forecasts at 25 % leads to R^2_{OOS} of 0.32 %, 0.56 % and 0.81 %, respectively, where the RF120 and LASSO120 are now significant. For LGBM120, results are more mixed as R^2_{OOS} for LGBM120 rises from 1.4 % to 1.6 % using winsorization of 0.05 % and then declines back to 1.4 % with a winsorization of 20 %. Winsorization has little effect on the longer windows. Winsorizing ensemble forecasts increases their predictability by approximately 20 % on average from 0.9 % to 1.1 %. The last three columns present different levels of shrinkage towards the mean. Shrinkage effects differ across methods, leading to modest improvements in LGBM120, ENet120 and XGB120, but limited effects on ML methods using a longer window. Shrinkage lowers ensemble predictability as pooling already reduces the variation so shrinkage is likely unnecessary.

Fig. 2 presents the optimal winsorization and shrinkage parameters when calculated using two halves of the sample using forecasts for the ten-year rolling window. For most ML methods, the optimal levels of winsorization and shrinkage are relatively high, and vary over the sample. The only procedure that does not require winsorizing forecasts is the recursive SCF in the upper left quadrant; instead, during one period, shrinking towards the HA by 10 % promotes performance. The only ML method that requires little winsorization and shrinkage is the SCF120 for the first half; however, during the second half, for optimal predictability, forecasts need 10 % winsorization. In contrast, LASSO120 requires 30 % and 50 % winsorization over the two samples, and no shrinkage; 50 % winsorization implies the median is optimal. For optimal predictability, LGBM requires no shrinkage for both samples, and no winsorization for the second half, but 50 % for the first half. ENet, XGB and NN require varying degrees of both shrinkage and winsorization over both halves of the sample. Overall, these figures indicate that shrinkage and winsorizing forecasts often are important for raising predictability but their degree varies over time. The results confirm that ML models are prone to large forecast failures due to overfitting or parameter instability.

Table 4 compares how winsorizing forecasts affect the large ML forecasts and squared forecast errors. Columns I (II) present the ratio of the squared forecast errors of the ML forecasts relative to the actual return with no winsorization to the squared forecast errors of the ML forecasts relative to the actual return with 10 % winsorization for the average of the top ten observations (ten%). Column I shows the top ten squared errors for SCF, NN and XGB are over 1000 times larger for the nonwinsorized procedure; the SCF particularly has very large squared forecast errors relative to its winsorized forecasts. Columns III and IV present the same ratio using the average of the top 20 and 20 % winsorization, and indicate that this method substantially lowers squared forecast errors for extreme observations. Ratios are higher for the ten (twenty) highest observations than the top 10 % (20 %), since winsorization mitigates extreme observations. Results support earlier evidence that the shorter 120 horizon has larger extreme observations than the 240 window. For conciseness, we report only the mean ensemble method and results show it has less extreme forecasts than the 120 horizon; the ensemble method however has larger squared errors than the longer ML horizons, except for the NN.

Table 3
R^2_{OOS} comparison of winsorization and shrinkage to the mean.

Model	Window	WZ0	WZ025	WZ05	WZ15	WZ20	WZ25	SHK.5	SHK.9	SHK.95
SCF	REC	0.037	−0.064	−0.075	−0.049	−0.013	0.011	0.262	0.121	0.082
SCF	120	−37.980	0.158 *	0.315 *	0.722 *	0.888 *	1.037 ***	−9.899	−30.909	−34.353
LASSO	120	−1.919	−1.023	−0.642	0.475	0.658	0.811 ***	−0.557	−1.582	−1.746
ENet	120	0.216 **	0.252 *	0.422 *	1.032 ***	1.320 ***	1.425 ***	0.645	0.388	0.307
LGBM	120	1.627 ***	1.574 ***	1.591 ***	1.356 ***	1.381 ***	1.328 ***	1.807 ***	1.822 ***	1.734 ***
RF	120	−2.236	−1.303	−1.396	−0.414	0.16	0.562 **	−0.103	−1.647	−1.932
XGB	120	0.033 ***	0.448 **	0.733 **	1.135 **	1.219 **	1.352 ***	0.522 **	0.212 **	0.127 **
NN	120	−8.442	−5.206	−3.503	−0.52	0.023	0.315	−2.504	−6.979	−7.693
SCF	240	0.041	0.084	−0.02	−0.023	0.019	0.071	0.189	0.098	0.071
LASSO	240	0.067 *	0.227 *	0.386 *	0.432 *	0.419 *	0.43	0.245	0.136	0.104
ENet	240	−0.634	−0.539	−0.502	−0.165	−0.147	−0.034	−0.129	−0.503	−0.567
LGBM	240	−0.059	−0.208	−0.22	−0.074	−0.049	0.018	0.217	0.035	−0.009
RF	240	−3.056	−2.493	−2.399	−1.842	−1.58	−1.25	−0.865	−2.512	−2.777
XGB	240	−0.157	−0.156	−0.19	−0.057	−0.007	0.161	0.313 *	0.002	−0.074
NN	240	−1.451	−0.279	−0.088	0.117	0.167	0.267	0.227 *	−0.963	−1.198
ENSMEAN		0.937 ***	0.95 ***	0.978 ***	1.054 ***	1.095 ***	1.065 ***	0.55 ***	0.873 ***	0.906 ***
ENSMED		0.876 ***	0.889 ***	0.924 ***	1.006 ***	1.021 ***	1.028 **	0.509 **	0.814 **	0.846 **
ENSTMEAN		0.964 ***	0.989 ***	1.006 ***	1.062 ***	1.094 ***	1.085 ***	0.559 ***	0.895 ***	0.930 ***

The table shows the effects of winsorizing forecasts and shrinkage towards the mean at different levels. WZ0, WZ025, WZ05, WZ15, WZ20,WZ25 have the forecasts winsorized at 0 %, 2.5 %, 5 %, 10 %, 20 % and 25 %. SHK 0.5, SHK 0.9 and SHK 0.95 represent a 50 %, 10 % and 5 % shrinkage towards the mean. * indicate that the reduction in MSFE is significant according to Clark-West tests.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

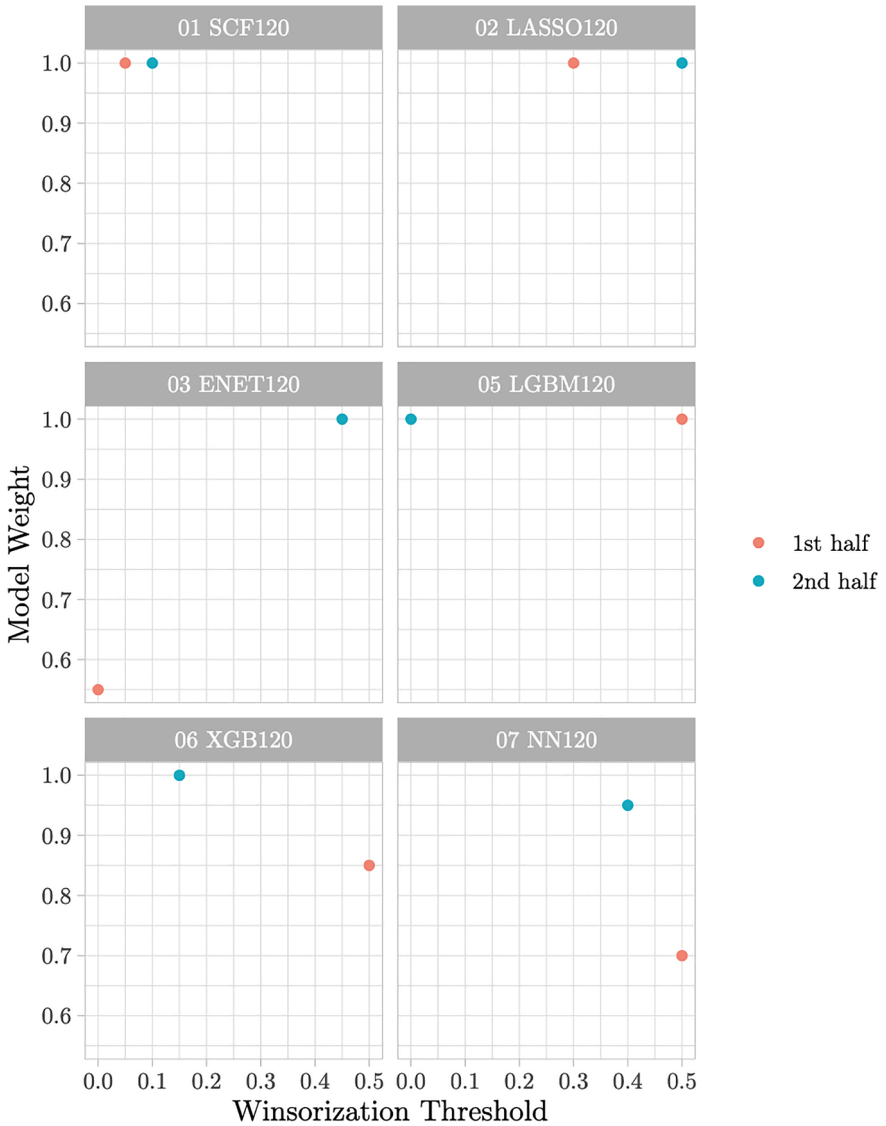


Fig. 2. Optimum winsorization and shrinkage first half and last half.
The graph compares the optimal winsorization and shrinkage levels of the first half and second half of the USA dataset, based on the optimized $R^2_{OOS}$for six of the base methods using a 120 window.

We illustrate in Appendix III the effects of 10 % winsorization by presenting two boxplots of the forecasts for our different methods using the rolling 120-month rolling window. Two figures are necessary due to the scale differing substantially. The unwinsorized data have numerous observations considerably outside the 1.5 interquartile range (IQR); in contrast, the ML procedures winsorized data have few data points out above the 1.5 IQR, implying winsorization shrinks the forecast.

Columns V-VIII indicate that winsorizing forecasts increase predictability by reducing extreme forecasts. The columns present the ratio of the ML forecasts to winsorized ML forecasts and indicate traditional ML forecasts are much larger than their winsorized counterparts. Column V shows that the top ten SCF forecasts are over 1,500 times more negative than the winsorized SCF forecasts, leading to the large squared forecast errors (in Columns I:IV). Results for the 120-month window reveal an interesting asymmetry of the forecasts that squaring hides. The ratios are one for most of the large negative forecasts, implying winsorization has no impact; however, the ratios are several times higher on the positive side. This indicates that large ML forecasts are all positive and require winsorization to lower their extreme forecasts. Appendix II uses this same analysis for the Goyal–Welch data, confirming that winsorizing substantially lowers the forecasts and forecast errors. Comparisons between tables reveal that winsorization lowers forecasts and forecast errors in similar magnitudes, hence winsorization success works for both high and low dimensional data. Several ML methods also possess substantial asymmetries of forecasts.

Table 4
Impact of winsorization on USA model results.

Model	Window	MLFE2/WMLFE2				MLF/WMLF			
		win10	win10 %	win20	win20 %	bot10	bot10 %	top10	top10 %
SCF	120	10,132.4	1747.9	375.5	65.8	−1,532.7	−305.9	101.4	18.9
LASSO	120	24.3	5.4	27.7	6.2	1.0	1.0	8.6	3.3
ENet	120	679.9	118.6	173.8	31.8	−3.4	0.1	24.7	8.0
LGBM	120	75.5	14.3	214,104.1	36,916.8	1.0	1.0	7.4	3.5
RF	120	235.1	41.8	5,551.5	959.1	1.0	1.0	3.9	2.1
XGB	120	1,807.1	312.7	18.9	5.0	1.0	1.0	6.2	3.0
NN	120	4,619.9	798.8	399.8	73.0	1.0	1.0	7.1	4.0
SCF	240	9.8	2.6	54.6	10.4	−6.8	−0.6	3.8	1.7
LASSO	240	23.8	5.2	15.2	3.9	−3.2	−0.7	5.8	2.2
ENet	240	8.7	2.5	92.6	17.2	−8.6	−1.3	5.4	2.3
LGBM	240	14.2	3.5	87.5	16.4	−14.1	−4.1	3.0	1.8
RF	240	5.7	1.9	69.6	13.3	0.7	0.9	3.4	1.7
XGB	240	35.7	7.2	134.9	24.7	1.0	1.0	6.2	2.7
NN	240	2,942.7	508.7	366,328.6	63,162.7	−45.9	−8.4	15.6	4.5
ENSMEAN		36.0	7.3	31.2	6.7	−4.6	−0.8	8.4	2.8

The table reports the results of each model under different winsorization thresholds. MLFE2/WMLFE2 is the ratio of the Machine Learning Forecast Error Squared (MLFE2) to the Winsorized Machine Learning Forecast Error Squared (WMLFE2). MLF/WMLF is the ratio of the Machine Learning Forecasts (MLF) to the Winsorized Machine Learning Forecasts (WML). Large ratios indicate that the winsorized forecasts reduced squared forecasts errors or forecast extremes. win10 and win10 % are the average of the top ten largest and 10 % winsorized; win20 and win20 % are the average of the top twenty largest and 20 % winsorized. ENSMEAN is the ensemble mean forecast. For instance, the ratio of the top ten unwinsorized squared errors (MLFE2) to the top ten squared residuals (MLWFE2) for the SCF is 10132; this indicates that SCF120 has extreme outlier forecasts (shown also in column 5 with a ratio of −1,537) that lead to high squared errors that winsorized forecasts.

Does forecast winsorization increase the R^2_{OOS} in other economies, and what is the optimal level of winsorization? Fig. 3 illustrates an inverse U-shaped relationship, peaking at approximately 20 % for CHN, GER and UK, 25 % for the US and 40 % for Canada. The tables and figures document several interesting findings. First, winsorizing forecasts substantially reduces both very large forecasts and squared forecast errors relative to the actual return. Second, winsorizing forecasts are more important at the ten-year horizon. Third,

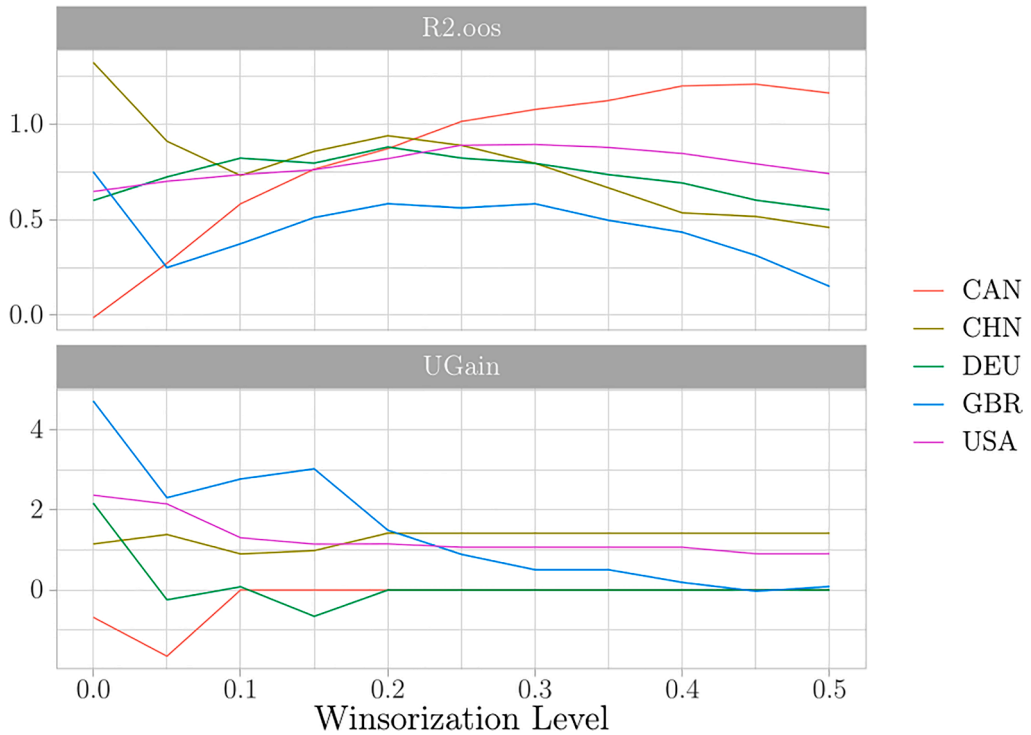


Fig. 3. Ensemble performance by level of winsorization.
The graph shows the 10 % trimmed mean results at different level of input model winsorization. The top panel shows the R^2_{OOS}value, and the bottom panel the Utility Gain. The lines were calculated using data from the full out-of-sample period.

large ML forecasts are asymmetric, and winsorization effectively reduces them. Fourth, winsorizing forecasts improves predictability for both high and low-dimensional data. Lastly, winsorized forecasts improve predictability in other economies, and ML forecasts require relatively high rates of winsorization.

3.3. Pooling forecasts

Different pooling or ensemble methods suggested by the literature can be structured broadly into: (a) methods that estimate the weights of the forecasts, (b) simple averaging methods, such as the mean, median, or trimmed mean of forecasts, and (c) methods based on prior model performance. Forecast combination began with Bates and Granger (1969), followed by Granger and Ramanathan (1984) and Diebold and Pauly (1987), who use regressions to estimate weights and hence are impractical given our high-dimensional data. These methods require inverting a large matrix, estimating a large number of parameters, and need a long time series window that exceeds the cross-section. In empirical work, estimation error and overfitting have led to poor results and simpler methods work better (Rapach and Strauss, 2008). Simple combination methods have surprisingly performed well, leading Stock and Watson (2004) to coin the phrase 'forecast combination puzzle.' Work by Clemen (1989), Smith and Wallis (2009) and Claeskens et al. (2016) provides additional insight. We also apply a DMA method (Raftery et al., 2010) that recursively adjusts predictions based on past model performance, where the coefficient forgetting factor of $\delta = 0.95$, model forgetting factor of $\alpha = 0.99$ and an error variance forgetting factor of $\beta = 0.95$. (Stock and Watson, 2004) introduce a historical inverse MSFE based weighting of forecasts (Stock and Watson, 2004), where we use a discount weight of 0.95.

Table 5 presents alternative ensemble specifications. The DMA procedure leads to negative R^2_{OOS} for all three window sizes; alternative forgetting factor specifications were tried with qualitatively similar results. Stock and Watson's (2004) method also generates negative R^2_{OOS} for all three window horizons; alternative discount weights of 0.90, 0.99 and 1 (no discount) lead to poor predictability. Although both methods are designed to lessen parameter instability and model uncertainty by weighting models with recent performance, they are unable to catch pockets of predictability fast enough, or their parameter estimates are too unstable. Their estimation errors outweigh any gains in potentially picking the correct models to more heavily weight.

Table 5 further supports the robustness of simple mean, median, and trimmed mean ensemble forecasts since they consistently outperform the HA across both windows with positive and significant R^2_{OOS}. The shorter ensemble window particularly generates economically large R^2_{OOS} with winsorized forecasts; e.g., for the full OOS period, the mean ensemble increases from 0.47 % to 1.16 %, and for the ex-Covid period, 0.78 % to 1.31 %. The bottom three rows present an alternative ensemble specification that pools forecasts from only the six ML procedures and excludes the SCF. These R^2_{OOS} are higher than Table 2; e.g., for the full sample period, the mean ensemble R^2_{OOS} increases from 0.75 % to 0.82 % and 0.67 % to 0.83 % for the median. For 1975:1–2019:12, the mean ensemble R^2_{OOS} increases from 1.00 % to 1.16 % for the simple mean and 0.97 % to 1.16 % for the median.

Table 5 also presents alternative winsorization parameters. Results for 5 %, 10 %, 20 %, and 25 % exhibit no clear pattern; winsorization improves ensemble predictability, but the precise amount varies by window size and method. The table additionally presents returns, Sharpe ratio, and Utility gain for the models that are winsorized at 10 % excluding the Covid period. The ensemble methods, particularly for the short horizon, raise returns by over 3 % per month, and the Sharpe ratios rise from 0.38 for HA to 0.66. The ensemble Sharpe ratios for the longer window range increase to 0.50–0.53, implying meaningful economic gains. Lastly, for the ensemble methods that average ML methods across both horizons, Sharpe ratios are approximately 0.48. Utility gains for the mean and trimmed mean ensemble exceed 3 % for the 120-month window, and approximately 1.5 % for methods combining both windows.

3.4. Robust evidence of winsorizing and pooling forecasts

Does pooling forecasts provide robust predictability in other economies? We first consider Canada, which has the second longest time series (beginning in 1984:07) and the largest number of variables (358). Due to a smaller time series, we consider a ten-year window and a recursive window beginning at ten years (120+). Column I in Table 6 shows only the ensemble forecasts possess significant and economically large R^2_{OOS}. Winsorizing forecasts at 25 % leads to significant predictability for the SCF and the recursive SCF, raising the R^2_{OOS} to 0.65 % and 0.64 %, respectively. High levels of winsorization generate significant predictability for NN120, LASSO120+, ENet120 and ENet120+. Winsorization at the 25 % improves the predictability of the mean, median and trimmed mean from 0.93 %, 0.99 % and 0.91 % to 1.13 %, 1.21 % and 1.18 %. Comparisons across winsorization specifications indicate 0.025 % has little impact on R^2_{OOS}, and then predictability increases monotonically from 5 % to 25 %. The last column demonstrates that shrinkage to the mean does not boost predictability. Larger shrinkage percentages further lower the R^2_{OOS}, with results available upon request.

For the UK, data begins in 1992:02 with 308 variables. Results in Table 7 differ from Canada and the US, except for the consistency of ensemble predictability. RF120 and RF120+ lead to significant R^2_{OOS} over 2 % and 4 %, and winsorization lowers the R^2_{OOS}. Shrinkage to the mean for RF enhances forecasts in contrast to the ensemble outcomes; however, this shrinkage is sensitive to specification, and 20 %, 30 % and 50 % do not improve results, with results available upon request. In contrast, winsorizing forecasts substantially improve the accuracy of SCF and LGBM; the R^2_{OOS} is negative with no winsorization and peaks at 1.72 % for SCF with 0.025 % winsorization, and 1.23 % for LGBM with 20 %. The SCF result resembles the US; SCF has less estimation error, but there are hundreds of variables to estimate, and a few large outlier forecasts require winsorization. Shrinkage towards the mean is insufficient to reduce these large forecasts. Winsorization improves all three ensemble results, with the highest R^2_{OOS} occurring at 10 % winsorization.

Table 5
Ensemble results with winsorization levels and ex-Covid comparison.

Model	Window	2021W0	2021W10	2019W0	2019W10	2019W5	2019W20	2019W25	RET	SHR	UG
HA									6.141	0.378	
SCF	REC	0.017	−0.088	0.037	0.473	−0.07452	−0.077	0.011	8.054	0.462	0.048
EDMA	120	−8.064	−2.898	−8.62	−1.502	−4.163	−2.854	−1.098	5.624	0.410	−0.976
SW.120	120	−0.602	−0.601	−0.281	−0.299	−0.269	−0.274	−0.306	8.430	0.501	1.810
EDMA	240	−16.536	−15.809	−17.391	−18.353	−16.009	−16.6	−19.787	3.946	0.405	−1.134
SW	240	−1.066	−1.07	−0.729	−0.742	−0.729	−0.731	−0.749	8.088	0.465	1.062
EDMA	240	−15.911	−15.939	−16.898	−18.666	−16.259	−16.919	−20.001	3.339	0.325	−1.783
SW	240	−1.006	−1.01	−0.671	−0.686	−0.671	−0.673	−0.697	7.839	0.452	0.862
ENSMEAN	120	0.466 **	1.155 ***	0.777 **	1.276 ***	1.442 ***	1.393 ***	1.271 ***	9.375	0.657	3.440
ENSMED	120	0.248 *	0.882 ***	0.387 **	1.353 ***	0.831 ***	1.056 ***	1.439 ***	6.963	0.444	0.860
ENSTMEAN	120	0.466 **	1.155 ***	0.777 **	1.276 ***	1.442 ***	1.393 ***	1.271 ***	9.375	0.657	3.440
ENSMEAN	240	0.504 **	0.858 **	0.71 **	0.938 ***	1.239 ***	1.13 ***	1.062 ***	8.125	0.516	1.868
ENSMED	240	0.75 **	0.753 **	0.91 ***	0.984 ***	0.896 ***	0.925 ***	1.028 ***	8.546	0.530	2.124
ENSTMEAN	240	0.824 **	0.812 **	1.009	0.97 ***	1.038 ***	1.026 ***	1.085 ***	8.015	0.500	1.687
EMLMEAN		0.824 **	0.932 ***	1.027 ***	1.145 ***	1.096 ***	1.155 ***	1.103 ***	8.180	0.476	1.495
EMLMED		0.837 **	0.966 ***	1.016 ***	1.185 ***	1.101 ***	1.156 ***	1.132 ***	8.248	0.478	1.528
EMLTMEAN		0.825 **	0.921 ***	1.022 ***	1.156 ***	1.092 ***	1.131 ***	1.123 ***	8.166	0.472	1.446

The table shows R^2_{OOS} and economic gains of the ensemble models relative to the historical average. The ensemble methods are calculated using (1) the 120 observation training window of all methods including the ML methods and SCF, (2) 120 and 240 observation training windows for all methods and (3) 120 and 240 observation training windows for ML methods only, without the SCF. All ensembles apply a recursive 0.2 dual-sided winsorization to the base model forecasts. EDMA is dynamic model averaging (Raftery et al., 2010), SW uses the Stock and Watson (2004) inverse MSFE approach. RET, SHR and UG are the annualized return, Sharpe Ratio and Utility Gain. EMLMEAN, EMLMED and EMLTMEAN ensemble only the ML methods (does not include the SCF) and average across both 120 and 240 windows. * indicate that the reduction in MSFE is significant according to Clark-West statistics

*** $p < 0.01$

** $p < 0.05$.

* $p < 0.1$.

Table 6
Canada R^2_{OOS} results with winsorization levels.

Model	Window	WZ0	WZ025	WZ05	WZ10	WZ15	WZ20	WZ25	SHK10
SCF	REC	−0.14	−0.01	0.018	0.098	0.136	0.187	0.200	−0.50
SCF	120	−0.62	0.236 **	0.340 **	0.441 **	0.533 **	0.598 **	0.647 **	−1.11
LASSO	120	−0.62	0.236 **	0.340 **	0.441 **	0.533 **	0.598 **	0.647 **	−1.11
ENet	120	−1.34	−0.71 *	−0.46 *	−0.18 **	0.026 **	0.107 **	0.223 **	−3.37
LGBM	120	−3.89	−2.79	−2.04	−1.20	−0.73	−0.42	−0.12	−4.45
NN	120	−5.42	−2.43	−1.25	0.229	0.439	0.577 *	0.663 **	−1.82
RF	120	−2.36	−1.64	−0.86	−0.34	0.406	0.515 *	0.820 *	−0.91
XGB	120	−1.21	−0.48	−0.17	0.088	0.341 *	0.454 *	0.637 *	−0.13
SCF	120+	−0.16	0.000	−0.01	−0.10	0.097 *	0.285 **	0.342 **	−0.02
LASSO	120+	−0.16	0.000	−0.01	−0.10	0.097 *	0.285 **	0.342 **	−0.02
ENet	120+	−0.06	0.109	0.118	0.197	0.246 *	0.292 **	0.348 **	0.431
LGBM	120+	0.399	0.801	0.727	0.539	0.553	0.747 *	0.788 *	−10.7
NN	120+	−12.9	−7.74	−5.89	−1.64	−0.59	−0.37	−0.00	−3.37
RF	120+	−4.13	−3.38	−2.93	−1.10	−0.62	−0.58	−0.08	−2.10
XGB	120+	−2.46	−2.51	−2.58	−2.14	−1.83	−1.68	−1.57	0.850
ENSMEAN		0.930 **	0.942 **	0.936 **	1.020 ***	1.065 ***	1.093 ***	1.131 ***	0.899 **
ENSMED		0.986 **	0.968 **	0.961 **	1.018 ***	1.072 ***	1.120 ***	1.205 ***	0.827 **
ENSTMEAN		0.906 **	0.891 **	0.883 **	0.998 ***	1.080 ***	1.128 ***	1.179 ***	−2.35 **

The table presents R^2_{OOS} for different levels of forecast winsorization. * indicate that the reduction in MSFE is significant according to Clark-West tests. 120+ is a recursive window starting with 120 observations; a 240 window can not be used due to a limited OOS sample. SHK10 represents a 10 % shrinkage to the mean.
*** $p < 0.01$.
** $p < 0.05$.
* $p < 0.1$.

Table 7
UK R^2_{OOS} results with winsorization levels.

Model	Window	WZ0	WZ025	WZ05	WZ10	WZ15	WZ20	WZ25	SHK10
SCF	REC	−1.176	0.529	0.504	0.43	0.482	0.55	0.443	−0.885
SCF	120	−1.569	1.722 **	1.393 **	1.232 **	1.184 **	1.138 **	1.079 **	−1.042
LASSO	120	−0.009	0.46	0.423	0.425 *	0.421 *	0.396 *	0.374 **	0.024
ENet	120	−0.057	−0.016	0.12	0.418	0.446	0.66	0.658	0.054
LGBM	120	−0.586	0.248	1.040 *	1.144 **	1.163 **	1.123 **	1.002 **	−0.379
NN	120	−6.322	−4.797	−4.068	−3.094	−2.571	−2.317	−2.023	−5.299
RF	120	2.214 **	1.181 *	0.661	1.696 **	1.532 *	1.432 *	1.421 *	2.560 **
XGB	120	−1.915	−1.322	−0.953	−0.428	−0.168	−0.121	0.049	−1.49
SCF	120+	−1.176	0.529	0.504	0.43	0.482	0.55	0.443	−0.885
LASSO	120+	−0.414	−0.374	−0.329	−0.069	−0.219	−0.045	0.1	−0.25
ENet	120+	−0.882	−0.721	−0.543	−0.449	−0.375	−0.226	−0.117	−0.757
LGBM	120+	−0.839	−0.897	−0.81	−0.263	0.017	0.117	0.193	−0.724
NN	120+	−13.714	−11.561	−8.177	−4.648	−1.984	−0.4	0.609	−11.242
RF	120+	4.164 ***	3.285 **	1.904 *	1.880 *	2.089 *	1.917 *	1.631 *	4.398 **
XGB	120+	−2.273	−2.418	−2.405	−2.743	−1.908	−1.038	−0.778	−1.834
ENSMEAN		0.72	0.734 *	0.763 *	0.827 *	0.824 *	0.787 *	0.757 *	0.670 *
ENSMED		0.571	0.676 *	0.713 *	0.726 **	0.716 **	0.711 **	0.644 **	0.526 *
ENSTMEAN		0.886	0.920 *	0.974 *	0.989 **	0.991 **	0.961 **	0.866 **	0.818 *

The table presents R^2_{OOS} for different levels of forecast winsorization. * indicate that the reduction in MSFE is significant according to Clark-West statistics. 120+ is a recursive window starting with 120 observations. SHK10 represents a 10 % shrinkage to the mean.
*** $p < 0.01$.
** $p < 0.05$.
* $p < 0.1$.

Germany's data begins in 1994:05 and has 307 variables. Table 8 shows the simple median and trimmed mean ensemble are the only significant methods at 95 % in column I. Winsorization improves predictability for XGB120 and ENet120. LGBM is significant at high levels of winsorization for both the 120 and 120+ windows. Winsorization boosts predictability for the simple mean. Shrinkage towards the mean lowers predictability.

China has a shorter time series that begins in 2008:05 with 281 observations and is presented in Table 9. The shorter time series, coupled with the relatively large volatility of return and predictors, leads to volatile ML forecasts, large Clark-West critical values, and thus difficulty rejecting the null. The simple mean and trimmed mean R^2_{OOS} are economically large at approximately 1 %, but are only significant at 10 % for high winsorization levels.

Table 8
Germany R^2_{OOS} results with winsorization levels.

Model	Window	WZ0	WZ025	WZ05	WZ10	WZ15	WZ20	WZ25	SHK10
SCF	REC	0.254	0.115	0.218	0.136	0.241	0.215	0.264	0.234
SCF	120	0.279	0.190	0.111	−0.01	0.087	0.136	0.222	0.263
LASSO	120	0.279	0.190	0.111	−0.01	0.087	0.136	0.222	0.263
ENet	120	−0.68	−0.62	−0.50	−0.19	−0.21	0.010	0.129	−0.58
LGBM	120	−2.02	−0.64	−0.08	0.463	0.842 *	1.188 **	1.043 **	−1.56
NN	120	−0.38	−1.91	−1.48	−1.07	−0.62;	0.308	0.526	0.098
RF	120	0.415	0.859 *	0.498	0.735 *	0.858 *	0.872 *	0.885 *	0.529
XGB	120	−5.64	−2.67	−1.86	−0.74	−0.52	0.232	−0.03	−4.52
SCF	120+	0.254	0.115	0.218	0.136	0.241	0.215	0.264	0.234
LASSO	120+	−1.62	−0.28	−0.21	−0.10	0.119	0.271	0.334	−1.33
ENet	120+	0.319 *	0.466 **	0.476 **	0.459 **	0.436 *	0.390 *	0.375 *	0.310
LGBM	120+	0.167	0.921 *	0.980 **	0.737 *	0.777 *	0.797 *	0.952 **	0.426
NN	120+	−1.25	−2.15	−1.57	−1.02	−1.05	−0.76	−0.45	−0.88
RF	120+	−7.10	−6.59	−5.89	−5.52	−5.56	−5.51	−5.22	−5.82
XGB	120+	−3.69	−2.54	−2.05	−0.91	0.084	0.174	0.178	−3.01
ENSMEAN		0.558 *	0.624 *	0.550*	0.587 *	0.627 *	0.669 *	0.693 *	0.693 *
ENSMED		0.813 **	0.782 **	0.765**	0.691 ***	0.693 **	0.723 **	0.732 **	0.746 **
ENSTMEAN		0.796 **	0.766 **	0.720**	0.690 ***	0.713 **	0.734 **	0.738 **	0.738 **

The table presents R^2_{OOS} for different levels of forecast winsorization. * indicate that the reduction in MSFE is significant according to Clark-West statistics. SHK10 represents a 10 % shrinkage to the mean.
*** $p < 0.01$.
** $p < 0.05$.
* $p < 0.1$.

Table 9
China R^2_{OOS} results with winsorization levels.

Model	Window	WZ0	WZ025	WZ05	WZ10	WZ15	WZ20	WZ25	HA10
SCF	REC	−9.81	−5.88	−2.82	0.186	0.192	0.281	0.374	−8.13
SCF	120	−10.7	−6.27	−2.76	0.299	0.093	0.453	0.792	−8.91
LASSO	120	−10.7	−6.27	−2.76	0.299	0.093	0.453	0.792	−8.91
ENet	120	1.446	1.559	0.690	−0.01	0.276	0.229	0.268	1.419
LGBM	120	1.027 *	1.593 *	1.241 *	0.894	0.656	0.772	0.739	1.405
NN	120	1.139	0.680	0.712	0.478	0.991	1.461	1.806	1.388
RF	120	−4.29	−3.08	−1.62	−0.90	−0.13	−0.21	−0.18	−3.48
XGB	120	−31.9	−26.3	−22.0	−16.1	−13.3	−9.11	−4.93	−25.7
SCF	120+	−9.81	−5.88	−2.82	0.186	0.192	0.281	0.374	−8.13
LASSO	120+	−9.81	−5.88	−2.82	0.186	0.192	0.281	0.374	−8.13
ENet	120+	−1.20	−1.07	−1.22	−1.35	−0.10	−0.01	0.061	−0.98
LGBM	120+	0.246	0.529	0.216	−0.19	−0.50	−0.72	−0.95	0.418
NN	120+	1.002	1.112	0.917	0.859	0.423	1.380	1.783	1.304
RF	120+	−2.79	−2.42	−1.83	−1.17	−0.76	−0.44	−0.29	−2.39
XGB	120+	−12.5	−10.3	−8.20	−4.69	−2.37	−1.34	−1.02	−10.5
ENSMEAN		0.710	0.844	0.972 *	1.013 *	1.013 *	0.992 *	1.004 *	0.672
ENSMED		0.332	0.387	0.453	0.455	0.438	0.338	0.256	0.313
ENSTMEAN		0.9	0.965 *	1.018 *	0.968 *	0.889 *	0.868 *	0.805 *	0.831

The table presents R^2_{OOS} for different levels of forecast winsorization. * indicate that the reduction in MSFE is significant according to Clark-West statistics. SHK10 represents a 10 % shrinkage to the mean.
***$p < 0.01$.
**$p < 0.05$.
* $p < 0.1$.

Finally, we examine US returns using the Goyal–Welch dataset. Results in Table 10 column I show that ENet120 is the only individual ML method with significant predictability. However, the ensembles are also highly significant. Winsorization effectively bolsters predictability for the rolling ten and twenty-year methods from 0 % and 0.15 % to 1.16 % and 0.99 %. This confirms earlier results that the SCF can have large forecast errors, but a simple winsorization procedure significantly raises predictability. Winsorization boosts predictability for XGB using a rolling 240-month window. Winsorizing the mean, median and trimmed mean ensemble forecasts at 25 % increases the R^2_{OOS} from 1.10 %, 0.91 % and 1.07 % to 1.27 %, 1.13 % and 1.23 %; hence winsorizing forecasts also raises predictability for low-dimensional data like the Goyal–Welch data, as the ensemble R^2_{OOS} increases an average of 18 %. Shrinkage to the mean does not boost predictability.

Following Hao et al. (2020) and He et al. (2021), we ran the Huber loss function using the 120-month rolling window for LGBM, XGB and NN. Results in Appendix IV show the Huber loss function does not affect inference. The R^2_{OOS} for the MSFE (Huber) for the LGBM, XGB and NN are: 1.446 (0.331), 0.089 (−0.064) and −8.016 (−2.236). Other metrics such as mean absolute deviation and the

Table 10
Goyal–Welch R^2_{OOS} results with winsorization levels.

Model	Window	2021	2021WZ10	2019WZ0	2019WZ05	2019WZ10	2019WZ15	2019WZ20	2019WZ25	SHK10	RET	SRP
SCF	REC	0.168	0.179	0.269	0.318 *	0.292 *	0.2530	0.224	0.143	−0.293	12.688	0.822
SCF	120	−0.272	1.001 ***	0.613 **	0.983 ***	1.244 ***	1.411 ***	1.487 ***	1.177 ***	−0.126	12.688	0.822
LASSO	120	0.168	0.179	0.510 **	−0.479	0.008	0.357 *	0.609 **	0.831 **	−1.176	12.571	0.814
ENet	120	−0.187	0.781 ***	−2.853	0.781 **	0.934 ***	1.041 ***	1.108 ***	−0.314	−0.888	12.688	0.822
LGBM	120	−2.099	−0.688	−2.759	−0.946	−0.397	−0.01	0.363	0.246 **	−0.804	12.688	0.822
NN	120	−3.137	−1.571	−1.153	−2.68	−1.85	−1.328	−0.765	0.177	−12.102	12.127	0.819
RF	120	−5.632	−2.677	0.481 **	−2.998	−1.825	−1.066 **	−0.359	0.854 ***	−6.555	10.612	0.776
XGB	120	−1.421	−0.174	0.905 **	−0.624	−0.232	0.065	0.293 *	0.987 **	−1.301	12.688	0.822
SCF	240	0.001	0.807 ***	−0.299	0.587 ***	0.796 ***	0.898 ***	0.957 ***	0.517 ***	−0.065	12.688	0.822
LASSO	240	−0.272	0.122	−1.088	−0.216	−0.099	0.007	0.129	0.402	−4.299	12.688	0.822
ENet	240	0.911 **	1.179 ***	−2.42	1.14 ***	1.181 ***	1.172 ***	1.226 ***	0.311 *	−0.016	12.688	0.822
LGBM	240	−1.583	0.392 *	−2.22	−0.119	0.199	0.331 *	0.381 *	0.266 *	−0.581	12.688	0.822
NN	240	−3.158	−0.65	0.620 **	−1.64	−0.675	0.076	0.342	0.733 **	3.026	11.703	0.767
RF	240	−5.983	−1.762	0.796 **	−2.213	−1.241	−0.581	−0.162	0.875 **	−9.033	11.118	0.769
XGB	240	0.103 *	0.749 **	0.645 **	0.59 **	0.603 **	0.697 **	0.754 **	0.790 **	1.524	12.688	0.822
ENSMEAN		0.881 **	0.946 **	0.774 **	1.134 ***	1.162 ***	1.189 ***	1.217 ***	0.870 **	0.509	12.688	0.822
ENSMED		0.727 **	0.858 **	−1.232	0.994 ***	1.043 ***	1.07 *	1.092 **	0.326 *	0.645	12.688	0.822
ENSTMEAN		0.859 **	0.947 **	−0.077	1.121 ***	1.158 ***	1.194 ***	1.203 ***	0.208	0.548	12.688	0.822

The table presents R^2_{OOS} for different levels of forecast winsorization. The first two columns are the full sample, 1975–2021, and the remaining columns are 1975–2019, pre-Covid. * indicate that the reduction in MSFE is significant according to Clark-West statistics. SHK10 represents a 10 % shrinkage to the mean. RET represents annualized mean returns and SHP the Sharpe Ratio.

*** $p < 0.01$.

** $p < 0.05$.

* $p < 0.1$.

Pesaran-Timmermann probability test also exhibit only modest changes. A 10 % winsorization level further does not improve the R^2_{OOS} for the Huber loss function. Appendix V presents boxplots that compare the forecasts for LGBM, XGB and NN for unwinsorized, winsorized and Huber loss methods. For LGBM and NN, 10 % winsorization shrinks the forecasts considerably more than using the Huber loss; in contrast, for XGB, using Huber loss functions modestly shrinks the forecasts more than 10 % winsorization.

4. Deeper dive in pooling, sparsity and parameter instability

4.1. Explaining pooling success

Why do pooling forecasts improve predictability? Similar to simple combination forecasts, pooling diversifies forecasts and works well when different ML methods outperform the HA at varying times. Using our US high-dimensional data, we illustrate this concept

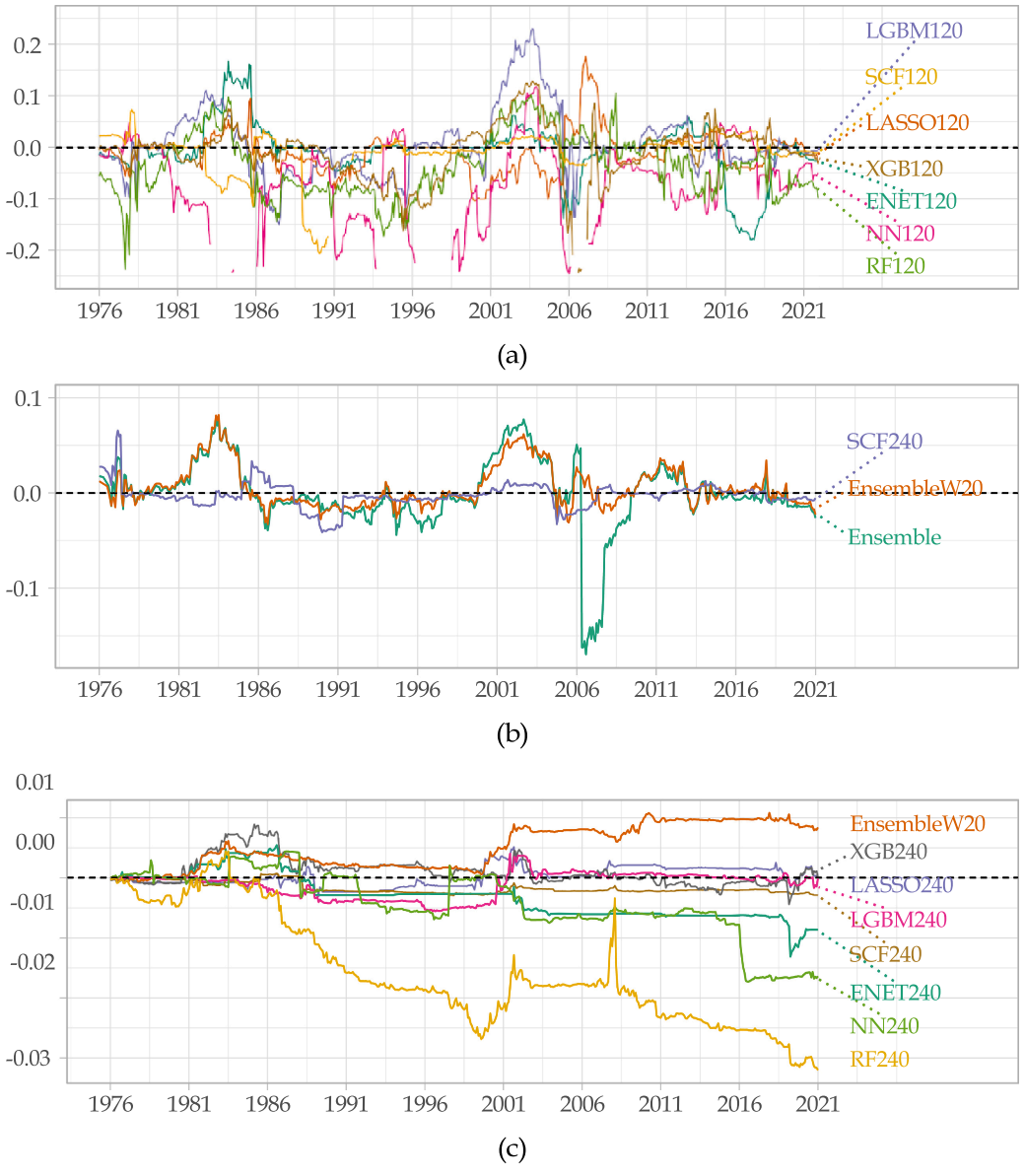


Fig. 4. Rolling R^2_{OOS} and Goyal–Welch difference in the CumSum.
The graph illustrates the lack of consistent performance over the sample of the ML methods. Panel (a) shows the 36-month rolling R^2_{OOS} of all the base models for the 120 ML models, panel (b) shows the 36-month rolling R^2_{OOS} of the Ensemble, and the winsorized Ensemble using a 20 % bi-sided winsorization of the base models (EnsembleW20) and (c) shows the cumulative sum of square differences between the ML240 base methods, 20 % winsorized Ensemble and the HA benchmark.

using rolling R^2_{OOS} in Fig. 4. Rolling R^2_{OOS} are a useful tool to highlight pockets of predictability and forecast failures. Panel (4a) illustrates the ML performance using a 120-month training window. All ML methods experience several episodic pockets of predictability with monthly large R^2_{OOS} that exceed 10 % in the mid-1980s and 2000s; however, what limits their full sample R^2_{OOS} shown in Table 2 is they all also experience large forecast breakdowns where negative R^2_{OOS} exceed 10 %. The best forecast performance, LGBM120, has one very good period in the mid-2000s, where predictability exceeds 20 %; however, this may be 'luck' as its performance before and after is lackluster.

Panel (4b) plots only the SCF240, ensemble and 20 % winsorized ensemble to contrast different methods of combining or pooling. The SCF performance differs considerably from the ML in Panel (4a), since the R^2_{OOS} variation is considerably more moderate, varying between ±2 % for most of the sample. We do not plot the SCF120 as its unwinsorized forecasts experience a very large forecast breakdown that dominates the plot. The ensemble forecast has three more periods of substantial predictability but one large forecast breakdown during the financial crisis. In contrast, the winsorized ensemble retains the pockets of predictability but avoids a large forecast breakdown and hence has superior performance than the unwinsorized ensemble. Hence, the rolling R^2_{OOS} plots effectively illustrate that winsorizing forecasts mitigate large forecast failures, supporting Table 4.

Panel (4c) presents Goyal Welch graphs for the 240-month window models. The 120-month window is similar but more volatile and not illustrated for brevity. The graph displays that most ML methods do not beat the HA; however, the winsorized mean ensemble method has several periods of sustained positive slopes that match the positive R^2_{OOS} in the panel above. The trimmed mean and median resemble the mean and are not plotted for conciseness. Because the winsorized ensemble does not have large periods of negative R^2_{OOS}, its Goyal–Welch plot slope is largely positive, implying that pooling ML models consistently leads to enhanced predictability.

To compare forecasting performance, we use the Giacomini-White pairwise comparison test (Giacomini and White, 2006), based on conditional predictive accuracy (CPA). This test expands on the Diebold-Mariano test (Diebold and Mariano, 1995) by accounting for parameter estimation uncertainty using the CPA. The null hypothesis is that both models have equal predictive ability, and rejection means that one model has significantly higher predictive accuracy than the other.

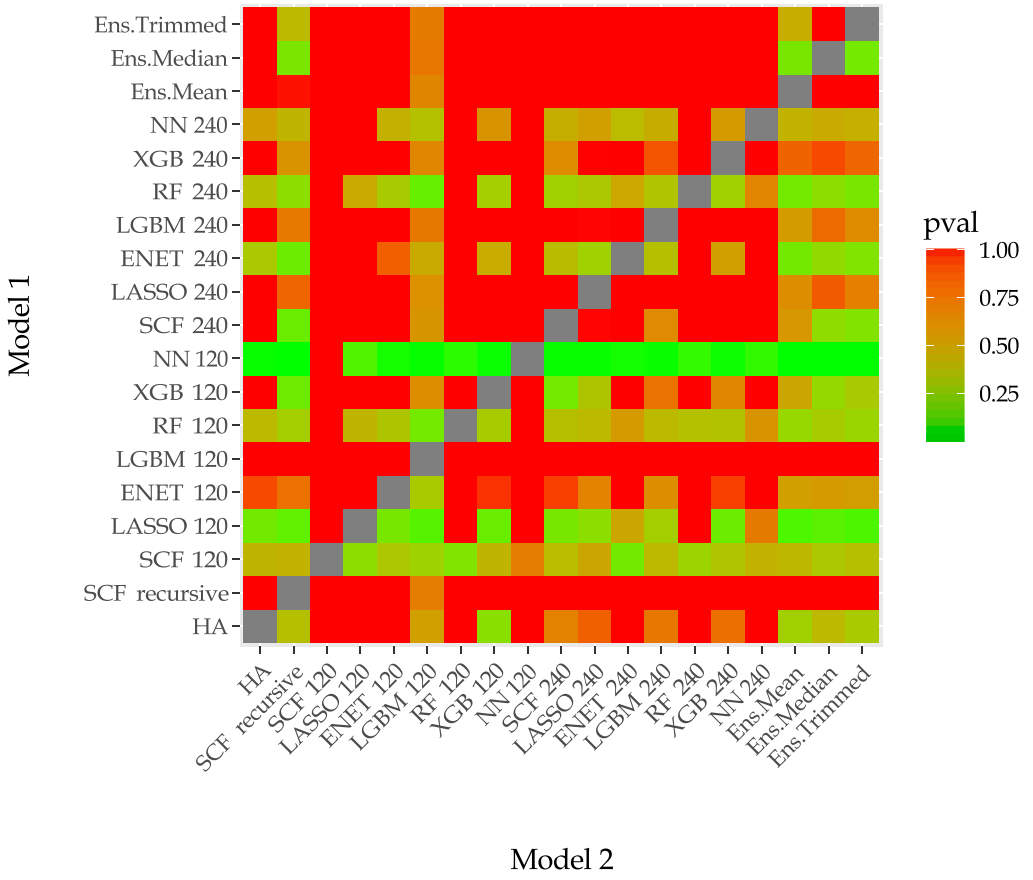


Fig. 5. Full sample relative method performance.
Each tile represents the p-value of the directional Giacomini-White test Giacomini and White (2006), of whether the method on the y-axis and the x-axis have the same accuracy. Rejection of the null indicates that the method on the y-axis has significantly higher Conditional Predictive Accuracy (CPA) than the method on the x-axis. The values on the tiles represent the p-values of the test.

Fig. 5 presents the tests, where a bright green result signifies a low p-value, which means that the method in the column forecasts significantly better than the method in the row. Examining the columns shows LGBM120, XGB120 and the recursive SCF significantly outperform (and hence are green) the other ML methods. The ensembles outperform all ML methods except these three methods, and have green in seven rows, highlighting that ensemble methods have higher predictive accuracy than most individual ML methods.

Instead of calculating the Giacomini and White (2006) test for the entire time frame, Fig. 6 shows a rolling comparison of 36-month windows. Panel (6a) compares shrinkage and selection to a simple linear combination forecast. Areas shaded in black represent periods when the method listed on the y-axis outperforms the SCF at the 95 % confidence level. While none of the methods is always significantly better than the SCF, selection with LASSO performs significantly better than the SCF 42.6 % of the time, which is more than pure shrinkage with Ridge, which is significantly better than SCF 29.6 % of the time. Most of the time that shrinkage outperforms the SCF, selection also outperforms the SCF, although there are some exceptions such as the late 1970′s and mid-1980s. ENet outperforms in periods where neither pure shrinkage nor pure selection performs better than the SCF, such as between 2000 and 2007, and significantly outperforms the SCF 50 % of the time. Thus, there is value in shrinkage and selection, and ENet, which combines both, has the strongest ability to outperform the simple linear combination of predictors over time.

Panel (6b) compares the CPA of the ML methods to the SCF and indicates that LGBM significantly outperforms SCF 57.4 % of the time, and XGB, RF, ENet and NN outperform SCF 55.6 %, 38.9 %, 50 % and 22.2 % respectively. The graph indicates a strong overlap in the periods where the ML methods forecast relatively well, and there are several periods, such as the mid-1980s to 1990, the mid-

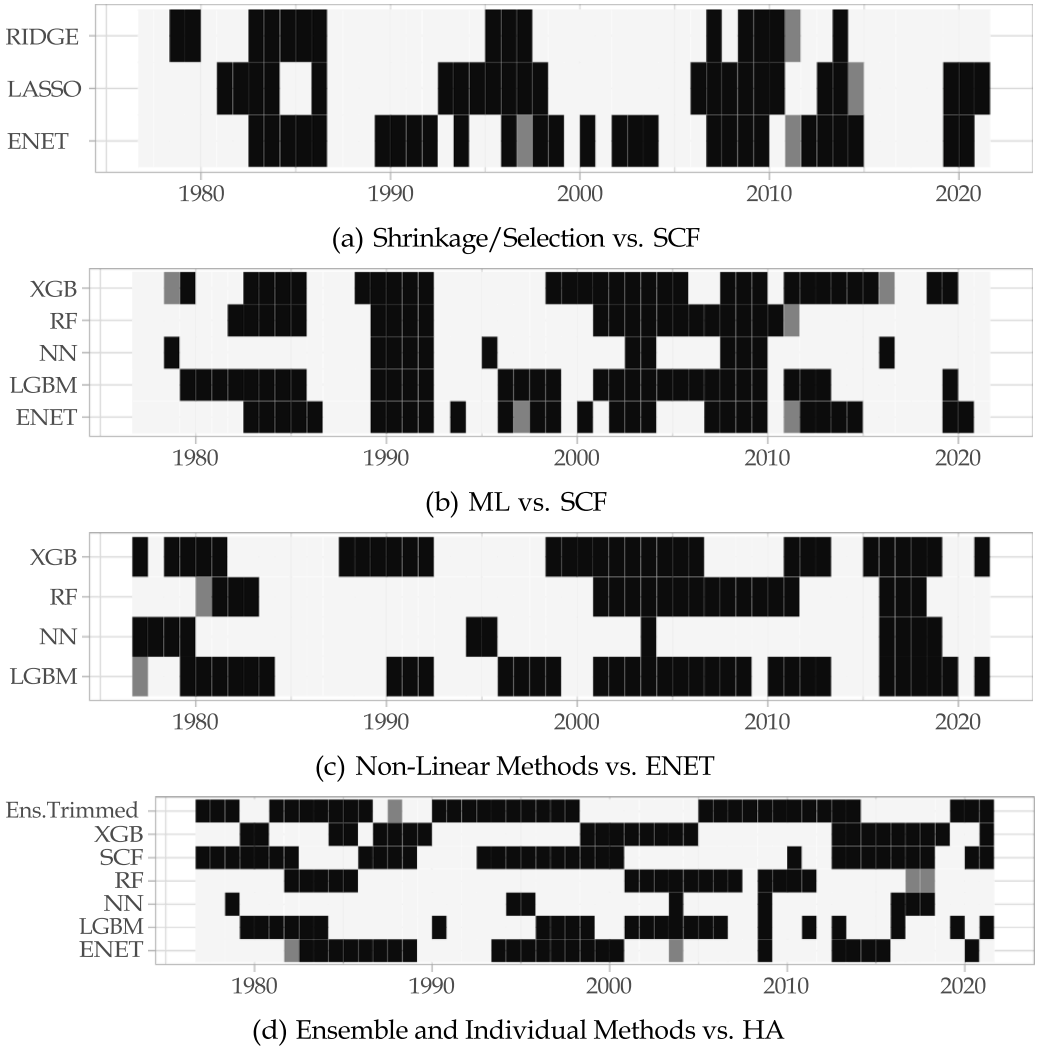


Fig. 6. Rolling relative method performance.
The graph shows the rolling 36 period 95 % significance (black) and 90 % significance (grey) that the method on the y-axis has a higher Conditional Predictive Accuracy (CPA) than the comparison method mentioned in the panel description, using the Giacomini and White (2006) procedure. Panel (a) compares LASSO, Ridge and ENet to SCF. Panel (b) compares the nonlinear ML methods and SCF, panel (c) compares the nonlinear methods to the linear ENet, and panel (d) compares the Ensemble procedure and several individual ML to the HA benchmark.

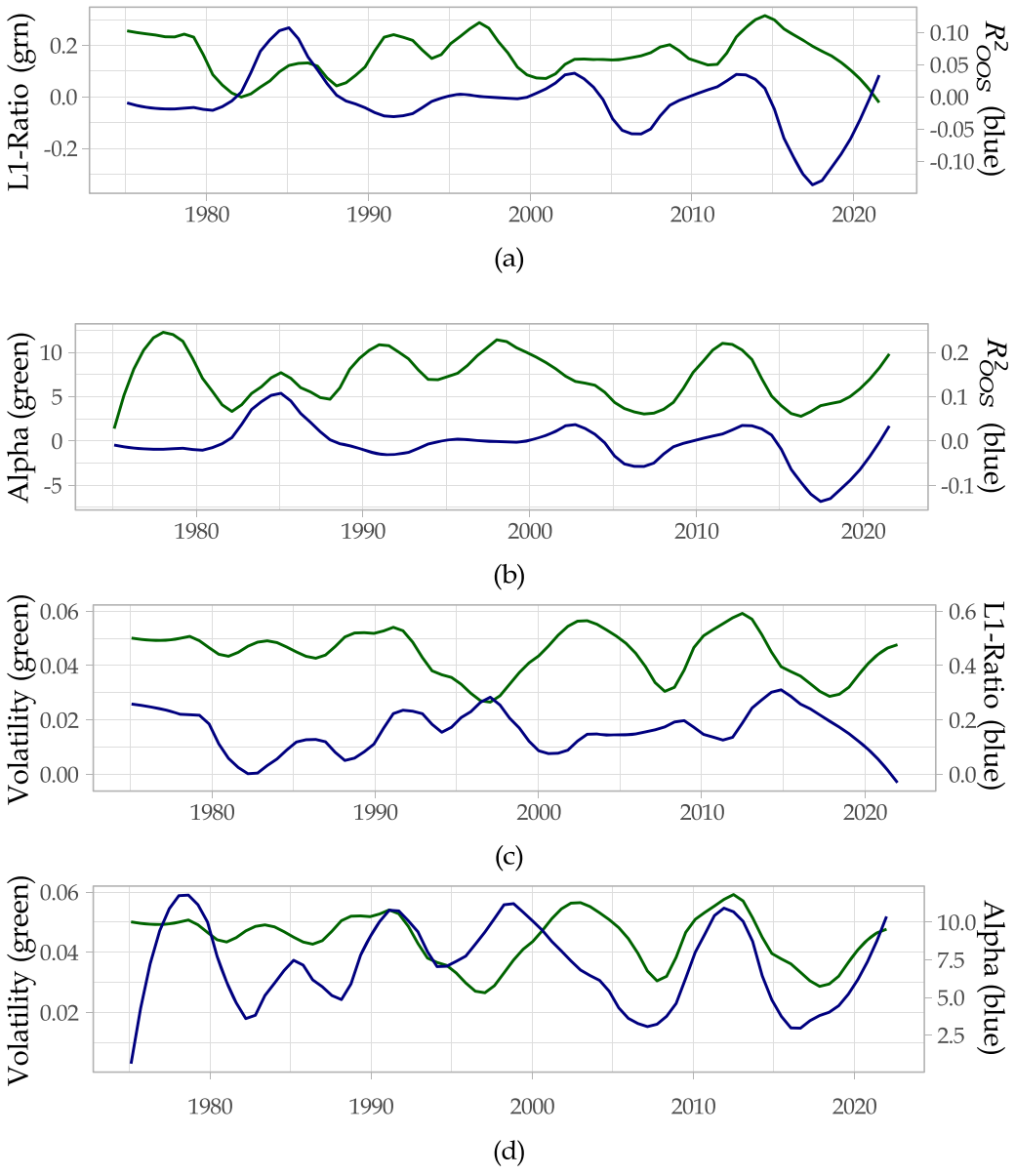


Fig. 7. Optimal parameters for ENet compared to model performance and volatility.
We compare optimal tuning parameters for ENet to model performance and external conditions (volatility) over time. This helps us investigate the conditions that determine choice of dense vs parsimonious parameter selection. Panel (a) shows the optimal ENet L1-Ratio parameter over time. If L1=1 (0), LASSO (Ridge) occurs. Panel (b) shows the rolling ENet shrinkage parameter *Alpha* compared to the rolling R^2_{OOS}. Panel (c) and (d) show the optimal ENet parameter values compared to volatility. Volatility is calculated based on a 60-month rolling window, rolling R^2_{OOS} is calculated based on a 36-month window.

1990s, and 2016–2018, where no ML method substantially outperforms the SCF. Panel (6c) compares the methods capable of modeling more complex relationships to ENet, which models pure linear relationships; note this is a high bar as ENet outperforms the SCF, LASSO and Ridge much of the time. LGBM significantly outperforms ENet 61.1 % of the time, while NN, RF and XGB significantly beat ENet 20.3 %, 35.2 % and 55.6 % of the time, respectively. This shows that there is value in considering non-linearities compared to linear methods much of the time.

Lastly, Panel (6d) illustrates the nonlinear methods and the trimmed ensemble against the HA to assess whether these methods consistently outperform the benchmark. The top row illustrates the trimmed ensemble has significantly better conditional forecast accuracy than the benchmark for most of the period from 1975 to 1999, 2007–2017, and then 2019–2021. All other nonlinear procedures also have periods of significant conditional accuracy, but their performance tends to be less consistent, and more resemble small pockets of predictability.

We also consider the model confidence set (MCS) of Hansen et al. (2011); however, this procedure lacked power and is unable to differentiate which model(s) are the best, likely due to instability over the sample. Appendix VI illustrates the MCS loss functions using a five-year rolling window, and shows ML loss functions and the relative performance substantially evolves over time. Several lines are discontinuous such as the HA, indicating the MCS drops it from the set due to its poor performance. For instance the historical average is missing for most of the first half of the sample (e.g., until 2002), implying the other ML outperformed it for over two decades.

Overall, ML methods have pockets of predictability followed by large periods of forecast failures, with large negative R^2_{OOS} and negative Goyal–Welch slopes. The forecast failures occur at different times, and no ML method outperforms the others for the entire out-of-sample period. Nonlinear methods perform well much of the time as compared to the simpler SCF. Pooling winsorized ML forecasts minimizes large forecast failures and frequently outperforms many of the ML models, providing consistent predictive accuracy over most of the sample.

4.2. Sparsity and volatility

We next investigate whether a sparse model performs better than a dense model, and then we examine environments that influence the performance of this relationship. The regularization parameters of ENet allow us to directly investigate the optimal level of shrinkage and selection based on data characteristics. The top two panels of Fig. 7 show a line for the optimal value of the *L1-Ratio* and *Alpha* parameters of ENet, respectively representing the weight of shrinkage (L1 close to 0) and selection (L1 close to 1), and the magnitude of shrinkage or selection (higher *Alpha* means more shrinkage or more stringent selection). These optimal values are determined through the tuning and cross-validation process described in Section 2.3. The graph suggests a positive relationship between both parameters and method performance. Table 11 shows the correlations between the optimal tuning parameters and R^2_{OOS}. The rolling R^2_{OOS} is significantly positively correlated with *Alpha* ($\rho = 0.12, p < 0.01$), implying predictability is higher when there is a higher magnitude of shrinkage/selection. Since the R^2_{OOS} is not significantly related to the *L1-Ratio* parameter, we can't conclude that either shrinkage or selection are mutually better.

Next, we evaluate factors that influence the parameters' model structure that determine the sparsity/density parameters. Return volatility, calculated as the 60-month rolling standard deviation of returns, is positively correlated with *Alpha* ($\rho = 0.17, p < 0.01$), implying higher volatility is associated with a stronger magnitude of shrinkage/selection. It is negatively correlated with the *L1-Ratio* ($\rho = -.15, p < 0.01$); hence, during periods of high volatility, shrinkage tends to be preferred over variable selection, indicating a more dense model predicts better. Volatility is strongly positively correlated with the rolling R^2_{OOS} ($\rho = 0.38, p < 0.01$), which suggests that ENet performance is higher during periods of higher volatility.

The positive relationship between predictability and volatility provides an interesting nuance to the last section's findings. LASSO outperforms the benchmark more often than Ridge because it successfully captures the relationship during normal, non-volatile times. However, the reverse is true during highly volatile times. ENet potentially outperforms both LASSO and Ridge over the OOS sample because it combines periods of rigorous selection with times of more lenient shrinkage.

4.3. Parameter instability and dimensionality reduction

Parameter instability implies the coefficient loadings on the independent variables are not static over time, often leading to poor forecast model performance (Rossi, 2021). To assess the effect on forecasting, we illustrate the variable importance over time, using the ENet coefficients as an example. Fig. 8 displays sharp fluctuations in variable importance, indicating pervasive instabilities. Parameter instability can be caused by structural breaks or an overlap in the definition of variables, particularly in high-dimensional spaces, where methods like ENet arbitrarily choose amongst a set of highly correlated predictors.

Several recent papers attempt to reduce large dimensional spaces by applying methods on top of the PC of the underlying data set (e.g. Kozak et al.,2020). Feng et al. (2020) indicate that this approach succeeds in specific settings but is statistically weaker. While PC

Table 11
Correlations between ENET tuning parameters, R^2_{OOS} and volatility.

Variable	Mean	Std.D	1	2	3
1. Alpha	7.20	5.23			
2. L1-Ratio	0.16	0.18	.28 ***		
3. R^2_{OOS}	−0.01	0.05	.12 ***	−0.07	
4. Volatility	0.04	0.01	.17 ***	−0.15 ***	.38 ***

The table presents correlations and significance between the ENET tuning parameters *Alpha* and *L1-Ratio* with 36-month rolling and 60-month rolling volatility. We find that the L1-Ratio positively correlates with Alpha, which indicates that a higher degree of selection goes along with higher magnitude of penalty. Further, higher model performance, as measured by R^2_{OOS}, correlates with a stronger penalty term, but does not correlate with the optimal shrinkage/selection. Lastly, a more volatile environment correlates with optimal models that are more dense (lower L1-Penalty) and have a larger magnitude of penalty (higher Alpha). The ENET model works better in periods of higher volatility, signified by the correlation between volatility and R^2_{OOS}.

*** $p < 0.01$
** $p < 0.05$
* $p < 0.1$.

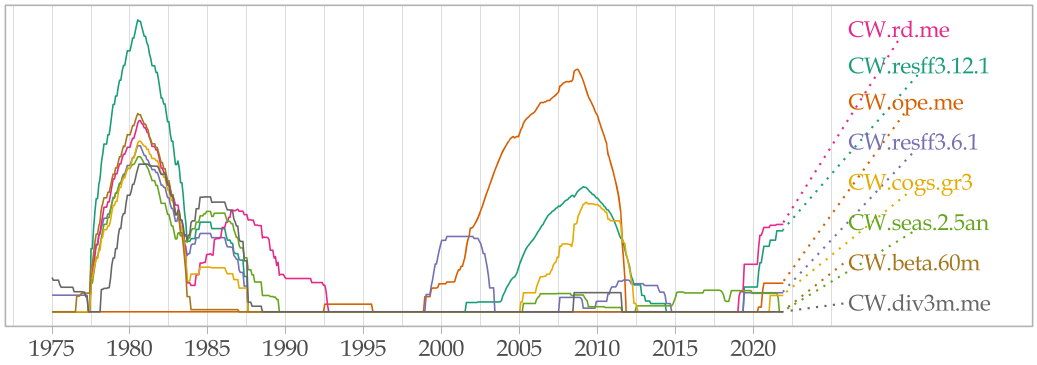


Fig. 8. Variable importance of Elastic Net.
The box shows the absolute variable importance of top-8 variables by aggregate importance in the =ENet 120-observations model and demonstrates strong instabilities in the underlying relationships over time.

reduce the dimensionality, they suffer from volatility in their composition over time. This occurs because a key assumption underlying these dimensionality reduction methods, constant factor loadings, is violated even over short periods such as a few years. While Stock and Watson (2004) show that PC can consistently estimate factors in the presence of small breaks, research by Chen et al. (2014) finds that ignoring breaks in the PC factor loadings leads to misleading inferences.

Fig. 9 illustrates the number of components it takes to explain cumulative percentages of the underlying variance. Panel (9a) shows that it takes 9, 61 and 132 components to reach 50 %, 90 % and 99 % of explained variance, respectively, when calculated post-hoc over the entire OOS period. However, in real-time forecasting, the variance of future observations is unknown, and hence, we only generate PC recursively using observations before the prediction time. The number of components required to reach 90 % of the variance on a rolling 60-month window of observations changes considerably over time. In 1960, five components are sufficient to describe 90 % of the variance of the independent variables, whereas in 2020, it takes at least 16 components to reach this level (Panel 9 (b)). The figure illustrates that when using a recursive approach, the number of PC grows steadily because the definition of the PCs, in terms of the underlying variables that constitute it, changes over time. Fig. 9(c) illustrates the rotation of the first PC for the first 20 variables of the underlying data and indicates stark differences in the definition of the PCs, which provides further evidence that considering parameter instability is paramount. Overall, evolving factor loadings introduce substantial estimation errors that negatively impact the forecasting performance, leading to non-significance of the PC methods; e.g., PC forecasts using a rolling window of 120 and 240 observations leads to R^2_{OOS} of -10.2 % and -15.2 %, respectively.

The supervised PLS procedure provides an alternative method to extract dense information by considering the best combination of independent variables to explain the maximum variance in the dependent variable. Fig. 10(a) shows that PLS requires 11, 99, 290 components to explain 25 %, 50 % and 54 % of the variance. Panel (10b) provides additional evidence that instability affects predictability is the maximum variance in the outcome explained by the number of components. A sharp drop-off in variance explained occurs for longer time frames since the relationship between variables evolves over time, making it difficult to summarize the variance succinctly over long periods. Fig. 10(c) shows the number of components required to describe 90 % of the variance for a ten-year window has fallen between 1970 and 2000, with a large drop in the late 1970′s. Fig. 10(d) illustrates factor loadings for the ten strongest variables within the first PLS component and clearly shows a high magnitude of instability. Overall, the results support the work of Li et al. (2018) and Fan et al. (2023) who caution against efforts to achieve sparsity.

Rolling windows can reduce parameter instability's impact by allowing coefficients to evolve over time. Because window sizes are arbitrary, window averaging may achieve a 'happy medium' of forecast window performance (Pesaran and Timmermann, 2007; Pesaran and Pick, 2011; Zhang et al., 2020). Table 12 presents model results for different window lengths of 60, 120 and 240 observations and a recursive training window with a minimum of 120 observations. Short training windows of 60 observations lead to stark model failure, likely due to parameter estimation error and overfitting in the training window. The recursive training window provides poor R^2_{OOS} since it does not account for evolving parameter estimates. This table supports our approach of combining ten and twenty-year windows; a smaller window provides imprecise estimation and fit, while a longer window does not allow the parameters to evolve over the sample.

Lastly, since we consider hundreds of predictors, we apply a rolling window of principal components to the base data and then use the ML models; note, PLS performance is a large order of magnitude poorer than PCA performance. Results in the last column reveal negative R^2_{OOS}, likely attributable to the component's time instability, where the composition of the components changes and the methods cannot effectively extract evolving information from the data. The poor predictability of the models using PCs compared to the original approach of using individual variables supports the findings of Li et al. (2018) and Fan et al. (2023) who 'embrace the blessing of dimensionality.' Thus, it is important to consider the instability of PC when using them as input variables for ML methods.

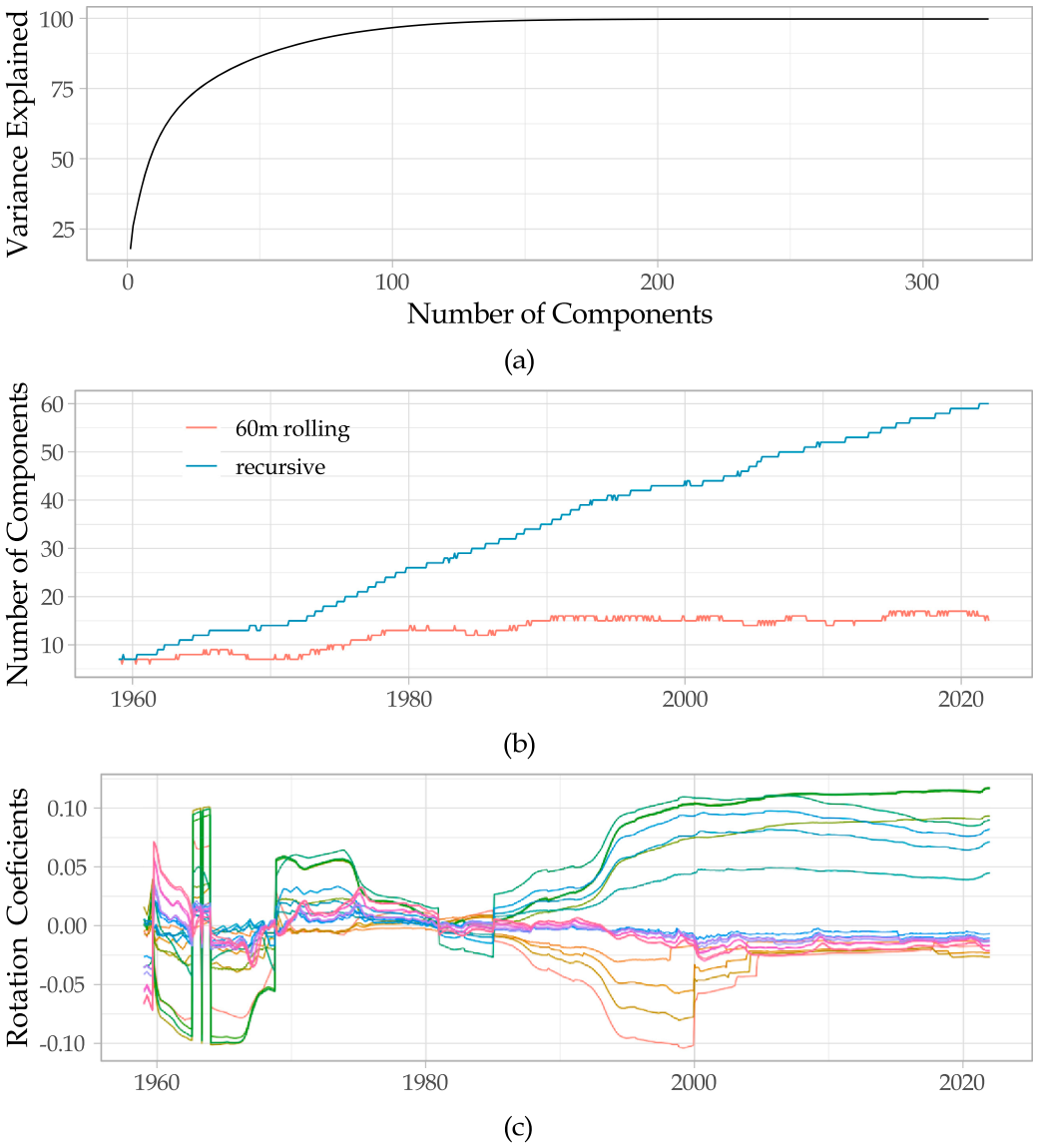


Fig. 9. Changing variance impact on principal components.
The graphs highlight the impact of changing variance in the independent variables on the construction of principal components. Panel (a) shows the cumulative percentage of variance in the underlying independent variables that is explained by the number of principal components on the x-axis using the entire out-of-sample data set. Panel (b) shows the number of components it takes to reach 90 % of variance explained for the USA data set when components are created in a 60-month rolling window and in a recursive fashion. The bottom panel (c) shows the rotational coefficients of the first 20 variables within the first principal component using a 60-month rolling window.

5. Conclusion

We evaluate the predictability of market returns in the US, Canada, China, Germany, and the UK using a novel high-dimensional data set of approximately 300 aggregated cross-sectional predictors per economy using pooled forecasts of machine learning models. Our ensemble method includes simple combination forecasts, LASSO, Ridge, Elastic Net, Random Forecast, Extreme Gradient Boosting and Light Gradient Boosting Machines. We observe that all machine learning methods have pockets of predictability, followed by periods of substantial model failure. Light Gradient Boosting Machine leads to significant predictability in the U.S. using a short ten-year rolling window but is highly sensitive to window specification, economies, and time period. Random Forest leads to a large R^2_{OOS} in the UK, ENet has significant predictability with a longer window in Germany and the Goyal Welch data, while no ML method forecasts well in Canada and China. Goyal–Welch graphs, rolling R^2_{OOS} plots, and rolling conditional predictive accuracy figures provide evidence that individual ML methods suffer from large forecast breakdowns and elusive predictability. Overall, driven by

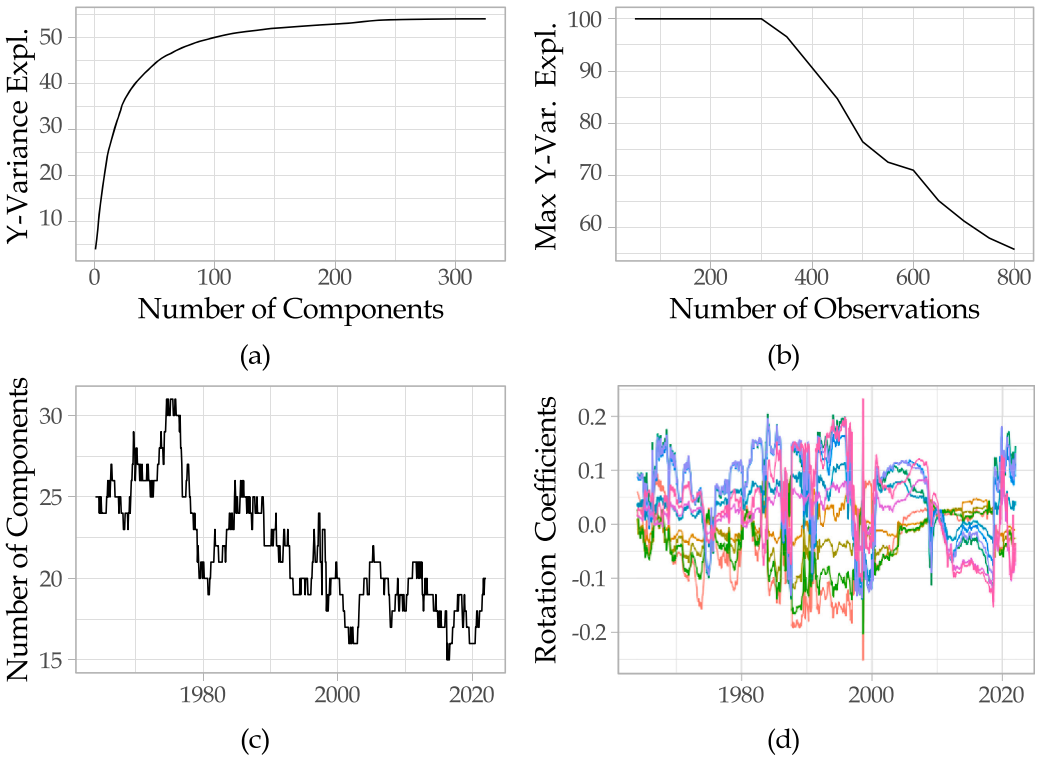


Fig. 10. Changing variance impact on PLS.
The graphs highlight the impact of changing variance in the independent variables on the construction of PLS. Panel (a) shows the cumulative percentage of variance in the underlying independent variables that is explained by the number of PLS components on the x-axis using the entire out-of-sample data set. Notably, this taps out at just over 50 % of the variance explained. Panel (b) shows that the cumulative variance explained in the Y-variable diminishes over a window of about 300 observations. Panel (c) shows the number of components it takes to reach 90 % of variance explained for the USA data set when components are created in a 120-month rolling window. Panel (d) shows the rotational coefficients of the first 10 variables within the first principal component using a 120-month rolling window.

Table 12
R^2_{OOS}Results for different rolling window lengths, recursive and PCA adjustment.

Method	Rolling 60	Rolling 120	Rolling 240	Recursive 120+	Rolling 120 PCA
SCF	−7.30	−33.07	0.02	0.05	−288.79
LASSO	−8.04	−1.86	0.05	−0.17	−1.71
ENET	−6.05	−0.04	−0.67	−0.84	−2.43
LGBM	0.21 **	1.43 ***	0.04	−0.11	−2.14
RF	−2.24	−3.16	−3.08	−1.74	−8.24
XGB	−2.73	0.15 **	0.01 *	−0.56	−0.66
NN	−21.04	−7.60	−1.65	−0.99	−3.37

The table presents the R^2_{OOS} statistics in percentages for the different rolling training window lengths, a recursive training window with a minimum of 120 observations and a rolling 120 window model with PCA adjustment. The number of PCs included in the model was determined based on a dynamic calculation of how many components are required to describe 99 % of the variance. Different thresholds for PC inclusion (90 % and 80 %) were tested with qualitatively similar results.
*** $p < 0.01$.
** $p < 0.05$.
* $p < 0.1$.

model uncertainty and parameter instability, no machine learning model consistently outperforms the benchmark across economies and the Goyal–Welch data that we consider for robustness.

Instead, pooling winsorized machine learning forecasts contribute to significant and economically large predictability across all five economies. Goyal–Welch graphs, rolling R^2_{OOS}, and rolling conditional predictive accuracy plots illustrate that winsorizing pooled forecasts consistently outperform the benchmark over time and do not experience large forecast breakdowns. Winsorizing forecasts substantially reduce large forecasts and forecast error, and outperforms a shrinkage to the mean approach because many machine learning predictions generate large errors that only winsorization can mitigate. Our analysis of sparsity and density questions the

practice of static variable selection, since a key assumption of constant factor loadings underlying these dimensionality reduction methods is violated even over short periods. Pervasive instabilities imply that principal components can not succinctly and dynamically capture the evolving density of the data, leading to poor out-of-sample predictions.

CRediT authorship contribution statement

Erik Mekelburg: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft. **Jack Strauss:** Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jempfin.2024.101538.

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